Tetrahedral Emergent Gravity vH3

The Minimal Simplex Principle and the Quaternion Projection Paradigm

From a Single Geometric Theorem to $\Omega_{\rm DM} = \frac{2}{3}\ln\frac{3}{2} \approx 0.2703$

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Abstract

Tetrahedral Emergent Gravity (TEG v8) derives galactic rotation curves from tetrahedral coordination $z_{\rm fund} = 4$ in $\mathbb{R}^3$ with zero free parameters. That coordination was a hypothesis. We identify it as a theorem by two independent routes, both anchored in two physical inputs stated explicitly as axioms.

Physical inputs (axioms). (i) The quantum vacuum is fundamentally described by the algebra of unit quaternions $\mathbb{H}$ (topologically S^3); all observed physics in $\mathbb{R}^3$ emerges as the natural projection $\pi : \mathbb{H} \to \mathbb{R}^3$ (Axiom 2.1). (ii) The 5-cell $\{3, 3, 3\}$ uniquely maximises holographic entropy density among all regular 4-polytopes at $R = 1$ (Proposition 3.1, established by direct computation on all six regular convex 4-polytopes; not derived from a dynamical principle — see Open Problem 2).

Derived results. Under these two axioms, the *Minimal Simplex Principle* — the combinatorial fact that the minimal convex polytope in $\mathbb{R}^d$ has $z(d) = d + 1$ facets — yields five theorems: (i) $z_{\rm fund} = z(3) = 4$ is a theorem, not a hypothesis; (ii) the holographic bit $\Delta S = \ln 2$ is universal; (iii) $d = 3$ is the unique dimension with integer bit count $N_{\rm bits} = d$; (iv) the ratio $D_V/D_A = 3/2$ is exact and unique to $d = 3$; (v) projection reduces coordination by exactly one.

A new identity unique to $d = 3$ connects three independent results:

$$z_{\rm pack}(\mathbb{R}^3) - z(3) \;=\; 12 - 4 \;=\; 8 \;=\; 2^{N_{\rm bits}(3)}.$$

This identity is a corollary of the factorisation $z_{\rm pack} = N_{\rm bits} \cdot z(3) = 3 \times 4$ (Proposition 5.2) combined with Theorem 4.4. The $\mathbb{Z}_2$ symmetry $a \mapsto -a$ of the S^3 sigma-model (Theorem 5.5) derives the factor 2 in the geometric frustration exactly. Together with the equipartition theorem of TEG v8, this yields the cosmological dark matter fraction without free parameters:

$$\Omega_{\rm DM} \;=\; \frac{2\ln(3/2)}{3} \;\approx\; 0.2703,$$

within 0.1% of the local SH0ES measurement and 4.4% of Planck 2018. The Newtonian limit is resolved by a chameleon mechanism with analytically derived transition

threshold $\rho/\rho_c \gtrsim 15.9$. A candidate for the baryon fraction is derived from the spectral entropy deficit of the 5-cell graph K_5: $\Omega_b = D_A - S_{K_5} \approx 0.0516$ (5.2% from observed).

Covariant extension (Appendix F). A covariant action for TEG is constructed by integrating the multi-fractional spacetime formalism of Calcagni (2012) with the tetrahedral axiom. The single free parameter of Calcagni's family is fixed exactly by the geometric axiom:

$$\alpha_{\mathrm{TEG}} = \frac{\ln 8}{4} = \frac{3\ln 2}{4}, \qquad D_{\mathrm{eff}} = 4\alpha_{\mathrm{TEG}} = \ln 8.$$

The resulting field equations reduce to standard Einstein equations sourced by matter and the scalar field $a(x)$; the multi-fractional measure cancels from the gravitational sector. General Relativity is recovered exactly in the high-density limit via the renormalised potential $\widetilde{V}(a,\rho) = V(a,\rho) - D_A$ with boundary condition $\widetilde{V}(0) = 0$. The fractal correction to the scalar equation of motion exists but is IR-suppressed by $(H_0/H_{\mathrm{Pl}})^2 \approx 10^{-122}$, leaving all main-text results unchanged. The holographic codimension $\partial = 3 - \ln 8$ acquires a first-principles geometric interpretation as $\partial = D_{\mathrm{spatial}} - D_{\mathrm{eff}}$. A conjectural Hubble constant $H_0^{\mathrm{TEG}} = H_{\mathrm{CMB}}/\sqrt{\partial} = 70.25\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Route A, Conjecture 3) is within 0.3σ of CCHP 2025 and 1.5σ of DESI 2024; an alternative structural route gives $61.8\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Route B), and reconciling the two constitutes the primary open sub-problem of the covariant extension.

Observational tests (Appendix D). Systematic tests against current public data are reported with honest assessments of discriminating power: the GWTC-3 second law (459 BBH events, 99.0% compliance); EHT shadow sizes for M87* and Sgr A* (consistent with the high-density GR limit); cluster lensing convergence for Abell 2744 ($\chi^2/N = 3.91$ vs. NFW 0.52, limited by 2011-era angular resolution); and the growth-rate parameter $f\sigma_8(z)$ from DESI DR1 ($\chi^2/N = 1.24$, zero free parameters, vs. Planck 0.95 with two parameters). The decisive near-term test is a direct Euclid Full Mission measurement of Ω_m: TEG predicts $\Omega_m^{\mathrm{TEG}} = 0.3193$, separable from Planck ΛCDM at $\sim 8\sigma$ with Euclid precision.

Scope of the derivation. All results from Section 4 onward follow by proved theorems from Axiom 2.1, Proposition 3.1, and the classical kissing number $z_{\mathrm{pack}}(\mathbb{R}^3) = 12$. The derivation of Ω_{DM} contains no additional conjectures beyond these three inputs.

Proposition 3.1 is established within TEG by direct computation but is not derived from a variational or dynamical principle; its deeper justification is identified as Open Problem 2. One open conjecture (Conjecture 3: $H_0 = H_{\mathrm{CMB}}/\sqrt{\partial}$) and five open problems are stated precisely: derivation of the baryon fraction from the causal asymmetry of the 5-cell (Open Problem 1); derivation of $V(a,\rho)$ from a single variational principle in the S^3 sigma-model (Open Problem 2); completion of the covariant formulation — Friedmann equation in FRW, derivation of $M_{\mathrm{vac}}(r)$ from $\widetilde{V}(a,\rho)$, and boundary condition $\widetilde{V}(0) = 0$ from a symmetry or renormalisation-group principle (Open Problem 3); joint derivation of H_0 and Ω_{DM} from the same geometric axiom to resolve the Planck–SH0ES discrepancy (Open Problem 4); and characterisation of all dimensions where the optimal-packing contact set decomposes as $z_{\mathrm{pack}}(\mathbb{R}^d) = d \cdot 2^{d-1}$, with $d = 3$ as the unique solution (Open Problem 5).

Note on the Current Status of the Manuscript (vH3)

This manuscript presents a work-in-progress version of Tetrahedral Emergent Gravity. While the core derivation of the cosmological dark matter fraction (Ω_{DM}) and the promotion of $z_{\mathrm{fund}} = 4$ from hypothesis to theorem are complete and rest on rigorously proved results, certain sections—particularly the appendices on black hole observational tests (Appendix E) and the thermodynamic structure of the S^3 sigma-model (Appendix C)—should be regarded as exploratory and preliminary documentation.

These appendices serve to record internal consistency checks and outline future predictions, but do not yet constitute comprehensive observational analyses. They are included to demonstrate the overall coherence of the framework and to guide subsequent development.

The author welcomes technical feedback, especially concerning the identified Open Problems and possible pathways toward a fully variational derivation of the vacuum potential.

Contents

1 Introduction and motivation

Tetrahedral Emergent Gravity (TEG v8, [1]) derives the effective vacuum dimension $D_{\text{eff}} = \ln 8$, the roughness parameter $\sigma_{\text{eff}} = 0.1088$, and a predictive rotation-curve equation from a single geometric axiom: the quantum vacuum in $\mathbb{R}^3$ selects tetrahedral coordination $z_{\text{fund}} = 4$ by maximising holographic entropy density among all Platonic solids. Applied to 171 SPARC galaxies [6] with zero fitted parameters, TEG v8 achieves RMSE $= 0.152\,\text{dex}$.

Two problems remained open: (1) the origin of $z_{\text{fund}} = 4$ was a hypothesis, not a theorem; (2) the cosmological dark matter fraction could not be predicted from first principles.

The present work resolves both. The central observation is:

> *Observed physics in $\mathbb{R}^3$ is the projection of a quaternion vacuum in $\mathbb{H}$. The information lost per node under $\pi : \mathbb{H} \to \mathbb{R}^3$ is exactly one holographic bit. The gravitational shadow of that lost bit is dark matter.*

The derivation has three layers:

1. The quaternion vacuum (Section 2) provides the physical mechanism and anchors the factor 2 in $D_V = \ln(2z_{\text{fund}})$ geometrically.
2. The Minimal Simplex Principle (Section 4) converts $z_{\text{fund}} = 4$ from hypothesis to theorem via five proved results.
3. A new combinatorial identity unique to $d = 3$ (Section 5) yields $\Omega_{\text{DM}} = 2\ln(3/2)/3$ under one stated conjecture.

Throughout, proved results are clearly distinguished from axioms, conjectures, and open problems. The derivation rests on two physical axioms (Axiom 2.1 and Proposition 3.1) and one classical theorem (the kissing number $z_{\text{pack}}(\mathbb{R}^3) = 12$, [3]). Every subsequent result follows by algebraic or combinatorial proof with no additional free parameters or fitted inputs. The chain is summarised in Section 8.

2 The quaternion vacuum and the projection $\mathbb{H} \to \mathbb{R}^3$

2.1 Axiom and projection

Definition 2.1 (Quaternion Vacuum Axiom)**.** The quantum vacuum is fundamentally described by the algebra of unit quaternions $\mathbb{H}$ (topologically S^3). All observed physics in $\mathbb{R}^3$ emerges as the natural projection $\pi : \mathbb{H} \to \mathbb{R}^3$.

A unit quaternion takes the form

$$q = a + b\,\mathbf{i} + c\,\mathbf{j} + d\,\mathbf{k} \in \mathbb{H}, \qquad a, b, c, d \in \mathbb{R}, \quad |q|^2 = 1.$$

The scalar part $a \in \mathbb{R}$ encodes the orientation of the vacuum node relative to $\mathbb{R}^3$; the vector part $\mathbf{v} = (b, c, d)$ is the projected observable. The projection discards a:

$$\pi : \mathbb{H} \to \mathbb{R}^3, \qquad q = a + \mathbf{v} \;\mapsto\; \mathbf{v}.$$

2.2 Exact information loss

Proposition 2.2 (Exact information loss)**.** *The projection* $\pi : \mathbb{H} \to \mathbb{R}^3$ *loses exactly one holographic bit per vacuum node:*

$$\Delta S \;=\; D_V - D_A \;=\; \ln(2z_{\text{fund}}) - \ln(z_{\text{fund}}) \;=\; \ln 2.$$

Proof. $D_V = \ln(2z_{\text{fund}})$ is the bulk entropy encoding the interior/exterior orientational duality of each simplex facet; $D_A = \ln(z_{\text{fund}})$ is the surface entropy. Their difference is $\ln 2$, independent of z_{fund}. □

Remark 2.3. The ratio $D_A/D_V = \ln 4/\ln 8 = 2/3$ is exact. It is the same ratio that produces $T_{\text{TEG}} = \frac{3}{2}T_H$ in the black-hole thermodynamics of TEG v8. Both are consequences of $z(3) = 4 = 2^2$, as proved by Theorem 4.5.

The factor 2 in $D_V = \ln(2z_{\text{fund}})$ is anchored by the interior/exterior orientational duality of simplex facets — a combinatorial property of convex polytopes, not a postulate. This is the physical anchor that justifies the frustration formula in Section 5.

3 The 5-cell as fundamental polytope in S^3

3.1 Holographic entropy density in S^3

The natural analogue of the TEG v8 entropy functional in S^3 is

$$\rho^{(4)}_{\text{holo}}(n) \;=\; \frac{\ln n}{V_\partial(n)},$$

where $V_\partial(n)$ is the total 3-volume of the boundary cells of the regular 4-polytope at hyperradius $R = 1$.

Proposition 3.1 (5-cell selection in S^3)**.** *The 5-cell* $\{3,3,3\}$ *uniquely maximises* $\rho^{(4)}_{\text{holo}}$ *among all six regular convex 4-polytopes at* $R = 1$*, with a margin of* $3.46\times$ *over the second-place 16-cell.*

Proof. Direct computation; see Table 1. □

Table 1: Holographic entropy densities for the six regular 4-polytopes at $R = 1$.

Polytope	n	$a(R{=}1)$	V_∂	$\ln n/V_\partial$
5-cell $\{3,3,3\}$	5	1.2649	1.193	**1.3496**
16-cell $\{3,3,4\}$	8	1.4142	5.333	0.3899
24-cell $\{3,4,3\}$	24	1.0000	11.314	0.2809
8-cell $\{4,3,3\}$	16	1.1547	12.317	0.2251
600-cell $\{3,3,5\}$	120	0.8740	47.214	0.1014
120-cell $\{5,3,3\}$	600	0.7136	334.22	0.0191

3.2 Robustness of the 5-cell selection under two independent entropy functionals

Section 3.1 selects the 5-cell using the boundary-entropy-density functional $\rho^{(4)}_{\text{holo}}(n) = \ln n/V_\partial(n)$, where $V_\partial(n)$ is the total 3-volume of the boundary cells. This is the four-dimensional analogue of the surface-entropy functional S/A used in TEG v8 (Eq. 1) to select the tetrahedron among the five Platonic solids in $\mathbb{R}^3$. It is worth noting explicitly that in $\mathbb{R}^3$, S/A alone does *not* select the tetrahedron uniquely: the tetrahedron and octahedron tie at $S/A = 0.3001$, and the tie is broken only by an auxiliary minimum-complexity criterion (TEG v8, Theorem 1). By contrast, S/V (entropy over solid 3-volume) selects the tetrahedron uniquely in $\mathbb{R}^3$, with no tie.

A natural question is therefore whether the 4-dimensional selection of the 5-cell is similarly sensitive to the choice of functional, or whether it is robust. We check this explicitly by evaluating the second natural candidate functional,

$$\rho^{(4)}_{\text{holo,V}}(n) = \frac{\ln n}{V_4(n)}, \tag{1}$$

where $V_4(n)$ is the 4-content (hypervolume) of the solid 4-polytope itself — the direct four-dimensional analogue of S/V, rather than S/A.

Table 2 reports both functionals side by side, using independently verified 4-content values (Coxeter, *Regular Polytopes*, 3rd ed., Table I) at unit circumradius $R = 1$.

Polytope	n	$V_4(R{=}1)$	$\ln n/V_4$	$V_\partial(R{=}1)$	$\ln n/V_\partial$
5-cell $\{3,3,3\}$	5	0.146	**11.024**	1.193	**1.3496**
16-cell $\{3,3,4\}$	8	0.667	3.118	5.333	0.3899
8-cell $\{4,3,3\}$	16	1.000	2.773	12.317	0.2251
24-cell $\{3,4,3\}$	24	2.000	1.589	11.314	0.2809
600-cell $\{3,3,5\}$	120	3.863	1.239	47.214	0.1014
120-cell $\{5,3,3\}$	600	4.193	1.526	334.22	0.0191

Table 2: Two independent entropy-density functionals evaluated on the six regular 4-polytopes at $R = 1$. Both select the 5-cell uniquely, with margins of 3.54× ($\ln n/V_4$, over the 16-cell) and 3.46× ($\ln n/V_\partial$, over the 16-cell, as in Proposition 3.1) respectively.

Result. Both functionals select the 5-cell uniquely, with comparable margins (3.54× and 3.46× respectively over the second-place 16-cell). This is a qualitatively different situation from the 3-dimensional case: in $\mathbb{R}^3$, S/A alone fails to select uniquely (the tetrahedron-octahedron tie), and an auxiliary criterion is required; in S^3, both natural 4-dimensional analogues of the entropy-density functional agree without any auxiliary tie-breaking. The 5-cell selection of Proposition 3.1 is therefore robust to this choice, in a way that the $\mathbb{R}^3$ tetrahedron selection of TEG v8 is not.

Honest caveat. This robustness check does not explain *why* the relevant functional changes character between $d = 3$ (where S/A ties and S/V breaks the tie) and $d = 4$ (where both S/V_∂ and S/V_4 agree). It also does not establish that $\rho^{(4)}_{\text{holo}}$ (boundary form, used in Proposition 3.1) and $\rho^{(4)}_{\text{holo,V}}$ (solid-content form, introduced here) are the same functional under any precise correspondence beyond producing the same maximizer; their absolute values and rankings of the remaining five polytopes differ (e.g. the 24-cell and 120-cell swap rank-4/rank-5 position between the two functionals). What this section establishes is narrower and fully justified: the specific claim that *the 5-cell maximizes holographic entropy density uniquely* does not depend on which of the two natural entropy-density functionals is used, strengthening Proposition 3.1 without resolving the deeper question of which functional (if either) is the physically correct one from first principles.

Reproducibility.

```
import numpy as np
data = {
    "5-cell":   {"n": 5,   "V4_R1": 0.146},
    "16-cell":  {"n": 8,   "V4_R1": 0.667},
    "8-cell":   {"n": 16,  "V4_R1": 1.000},
    "24-cell":  {"n": 24,  "V4_R1": 2.000},
    "600-cell": {"n": 120, "V4_R1": 3.863},
    "120-cell": {"n": 600, "V4_R1": 4.193},
}
for name, d in data.items():
    print(name, np.log(d["n"]) / d["V4_R1"])
# 5-cell gives the maximum: 11.0235, margin 3.54x over 16-cell (3.1176)
```

3.3 The projection theorem

The 5-cell has 5 vertices, each connecting to exactly 4 others: $z_{\text{fund}}(\mathbb{H}) = 4$.

Proposition 3.2 (Projection theorem)**.** *The stereographic projection $\pi : S^3 \to \mathbb{R}^3$ maps the 5-cell to the regular tetrahedron. Coordination is preserved ($z_{\text{fund}}(\mathbb{H}) = 4 \to z_{\text{fund}}(\mathbb{R}^3) = 4$); the polytope loses one vertex ($5 \to 4$), corresponding to the scalar component a of $q \in \mathbb{H}$.*

Therefore $z_{\text{fund}} = 4$ in $\mathbb{R}^3$ is a theorem of the quaternion geometry, not an axiom. The tetrahedron is the projection of the 5-cell.

Table 3: Quantities conserved and not conserved under $\pi : \mathbb{H} \to \mathbb{R}^3$.

Quantity	In $\mathbb{H}$	In $\mathbb{R}^3$	Conserved?
z_{fund}	4	4	Yes
N_{bits}	3 (exact)	3 (exact)	Yes
$D_V - D_A$	$\ln 2$	$\ln 2$	Yes
z_{pack}	24	12	No
σ_{UV}	0.1147	0.3263	No
σ_{eff}	0.0382	0.1088	No

Proposition 3.3 (Combinatorial equivalence of the two 5-cell projections)**.**

Combinatorially equivalent*: both map the 5-cell P_5 to the regular tetrahedron, both lose the same vertex v_0 (the pure-scalar state $a = 1$), and both preserve coordination $z_{\text{fund}} = 4$ as a graph invariant.*

Here $z_{\text{fund}} = 4$ is the facet count of the minimal simplex (Definition 4.2), not the graph degree of the projected skeleton. The edge to v_0 is the 'lost' facet whose information content becomes the holographic bit $\Delta S = \ln 2$ (Proposition 2.2).

Let $P_5 \subset S^3 \subset \mathbb{R}^4$ be the regular 5-cell inscribed in the unit 3-sphere, oriented so that one vertex coincides with the north pole:

$$v_0 = N = (1, 0, 0, 0) \in S^3.$$

The remaining four vertices v_1, v_2, v_3, v_4 are equidistant from v_0 and from each other, with geodesic angle $\arccos(-1/4)$ *in S^3.*

Two projections.

$$\pi_{\text{geo}} : S^3 \setminus \{v_0\} \to \mathbb{R}^3, \quad \pi_{\text{geo}}(a, b, c, d) = \frac{(b, c, d)}{1 - a},$$
$$\pi_{\text{alg}} : \mathbb{H} \to \mathbb{R}^3, \quad \pi_{\text{alg}}(q) = \mathbf{v} = (b, c, d) \text{ for } q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}.$$

Explicit coordinates. *By regularity of P_5, the condition $v_0 \cdot v_k = -1/4$ (inner product in $\mathbb{R}^4$) with $|v_k| = 1$ gives*

$$v_k = \left(-\tfrac{1}{4}, \tfrac{\sqrt{15}}{4}\,\mathbf{u}_k\right), \quad k = 1, 2, 3, 4,$$

where $\{\mathbf{u}_k\} \subset S^2 \subset \mathbb{R}^3$ are the vertices of a regular tetrahedron, satisfying $\mathbf{u}_j \cdot \mathbf{u}_k = -1/3$ for $j \neq k$.

Images of the four vertices.

$$\pi_{\text{geo}}(v_k) = \frac{\sqrt{15}}{5}\,\mathbf{u}_k,$$
$$\pi_{\text{alg}}(v_k) = \frac{\sqrt{15}}{4}\,\mathbf{u}_k.$$

Both images form a regular tetrahedron in $\mathbb{R}^3$, differing only by the scale factors $\sqrt{15}/5$ and $\sqrt{15}/4$ respectively.

The lost vertex. *Both projections lose the same vertex v_0:*

$$\pi_{\text{geo}}(v_0) = \infty \notin \mathbb{R}^3, \qquad \pi_{\text{alg}}(v_0) = \mathbf{0} \in \mathbb{R}^3.$$

In both cases v_0 is the unique vertex with pure scalar component: $a = 1$, $\mathbf{v} = \mathbf{0}$.

[Logarithmic scale-invariance of entropy loss] Let $\lambda > 0$ be any positive scale factor. The holographic entropy loss under projection is invariant under rescaling of the projected image:

$$\Delta S = D_V - D_A = \ln(2z_{\text{fund}}) - \ln(z_{\text{fund}}) = \ln 2,$$

independently of the metric scale of the projected tetrahedron. Consequently, the scale factor $\lambda = \sqrt{15}/4$ (under π_{alg}) versus $\lambda = \sqrt{15}/5$ (under π_{geo}) does not affect any TEG result.

Proof. All TEG observables derive from $D_A = \ln z_{\text{fund}}$, $D_V = \ln 2z_{\text{fund}}$, and their difference $\Delta S = \ln 2$. These depend only on the coordination number $z_{\text{fund}} = 4$, which is a combinatorial (graph) invariant of the tetrahedron. Under any rescaling $\mathbf{u}_k \mapsto \lambda\mathbf{u}_k$, the graph structure is preserved and z_{fund} is unchanged. Therefore ΔS, D_A, D_V, and all derived quantities (Ω_{DM}, $\mathcal{F}$, N_{bits}) are identical under both projections. □

Theorem 3.4 (Equivalence of projections for TEG)**.** *The algebraic projection π_{alg} and the stereographic projection π_{geo} are:*

[label=(iv)]

1. ***Metrically inequivalent**: their images differ by the scale factor $(\sqrt{15}/4)/(\sqrt{15}/5) = 5/4 \neq 1$.*

2. ***Combinatorially equivalent**: both map the 5-cell P_5 to the regular tetrahedron, both lose the same vertex v_0 (the pure-scalar state $a = 1$), and both preserve coordination $z_{\mathrm{fund}} = 4$ as a graph invariant.*

3. ***Physically equivalent for TEG**: by Lemma 3.3, the metric inequivalence is invisible to all TEG observables, which depend only on combinatorial structure.*

Proof. (i) follows from the explicit scale factors in Proposition 3.3. (ii) Both projections remove v_0 and map $\{v_1, v_2, v_3, v_4\}$ to the vertices of a regular tetrahedron proportional to $\{\mathbf{u}_k\}$. The graph of the 5-cell is K_5; removing v_0 gives K_4, the complete graph on 4 vertices, which is the 1-skeleton of the tetrahedron. This graph has $z = 3$ edges per vertex as an abstract graph, but each node retains $z_{\mathrm{fund}} = 4$ as its coordination number in the vacuum network (the edge to v_0 is the lost holographic bit $\Delta S = \ln 2$, not a removed physical link). (iii) follows from Lemma 3.3. □

Remark 3.5 (Physical interpretation of v_0). Under π_{geo}, the vertex v_0 is sent to infinity: it is the "point at infinity" of $\mathbb{R}^3$, representing a degree of freedom with no localised spatial image. Under π_{alg}, the same vertex maps to $\mathbf{0}$: its spatial projection is the origin, with no directional information. In both cases, v_0 encodes the scalar component a of the quaternion vacuum $q = a + \mathbf{v}$. Its removal under projection is the geometric realisation of the information loss $\Delta S = \ln 2$ per vacuum node (Proposition 2.2). The two descriptions are therefore dual representations of the same physical fact: *the scalar degree of freedom of the quaternion vacuum has no spatial image in $\mathbb{R}^3$, and its absence is dark matter.*

The equivalence of Theorem 3.4 holds for all TEG observables, which are constructed from entropy differences and combinatorial invariants. Metric quantities (angles, distances in the projected $\mathbb{R}^3$) are not observables at the current level of TEG; their inclusion would require a metric completion of the theory (Open Problem 3).

4 The Minimal Simplex Principle: five theorems

Theorem 4.1 (Minimal simplex coordination)**.** *The minimal convex polytope enclosing finite volume in $\mathbb{R}^d$ has exactly $z(d) = d + 1$ facets.*

Proof. A convex body in $\mathbb{R}^d$ requires at least $d + 1$ bounding hyperplanes. The d-simplex achieves this minimum with exactly $d + 1$ vertices and $d + 1$ facets. □

In particular $z(3) = 4$, so $z_{\mathrm{fund}} = 4$ is now a theorem.

Definition 4.2 (Entropy structure)**.** For a vacuum network whose nodes carry the minimal simplex in $\mathbb{R}^d$:

$$D_A(d) = \ln(d + 1),$$
$$D_V(d) = \ln 2(d + 1).$$

The factor 2 encodes the interior/exterior orientational duality of each facet.

Note on the logical status of the factor 2 in Definition 4.2. The factor 2 in $D_V(d) = \ln 2(d+1)$ is introduced here as a postulate, motivated by the interior/exterior orientational duality of each facet of the minimal simplex. It is used throughout Sections 2–4 in this form. We flag explicitly that this factor is later *identified* — in Remark 5.6, following the independent proof of Theorem 5.5 — with the $\mathbb{Z}_2$ symmetry $a \mapsto -a$ of the S^3 sigma-model, where it is derived rather than postulated. Readers are referred to Section 5.3 for that identification.

We note for clarity that this identification is currently established at the level of physical correspondence (the two hemispheres $a > 0$ and $a < 0$ are interpreted as the two orientational states of a frustrated facet), rather than by an explicit mathematical chain showing that the combinatorial property "each facet has an interior and an exterior side" and the analytic property "$V(a) = V(-a)$" are necessarily the same object. Both factors are numerically equal to 2 and the physical interpretation offered in Remark 5.6 is reasonable, but a fully rigorous derivation showing these are the same factor — rather than two independent quantities that coincide — has not been given. Establishing this rigorously, or stating clearly that they are taken as identified by physical interpretation alone, is a precise, well-posed addition to Open Problem 2 (derivation of the vacuum potential from first principles).

Theorem 4.3 (Universal holographic bit)**.** $\Delta S(d) = D_V(d) - D_A(d) = \ln 2$ *for all* $d \geq 1$.

Proof. $\Delta S = \ln 2(d+1) - \ln(d+1) = \ln 2$. □

Theorem 4.4 (Uniqueness of $d = 3$)**.** *The condition* $N_{\text{bits}}(d) \equiv D_V(d)/\ln 2 = d$ *has exactly one solution among positive integers:* $d = 3$.

Proof. $N_{\text{bits}} = d$ requires $d + 1 = 2^{d-1}$. Checking: $d = 1$: $2 = 1$ (false); $d = 2$: $3 = 2$ (false); $d = 3$: $4 = 4$ (true); $d \geq 4$: exponential beats linear. □

Table 4: Entropy structure by dimension. Only $d = 3$ gives $N_{\text{bits}} = d$.

d	$z(d)$	D_A	D_V	ΔS	N_{bits}
1	2	$\ln 2$	$\ln 4$	$\ln 2$	2.000
2	3	$\ln 3$	$\ln 6$	$\ln 2$	2.585
3	**4**	$\ln 4$	$\ln 8$	$\ln 2$	**3.000**
4	5	$\ln 5$	$\ln 10$	$\ln 2$	3.322
5	6	$\ln 6$	$\ln 12$	$\ln 2$	3.585

Theorem 4.5 (3/2 ratio uniqueness)**.** $D_V(d)/D_A(d) = 3/2$ *if and only if* $d = 3$.

Proof. $D_V/D_A = 1 + \ln 2/\ln(d+1) = 3/2$ iff $d + 1 = 4$, i.e. $d = 3$. □

Theorem 4.6 (Projection reduces coordination by one)**.** *Under* $\pi : \mathbb{R}^d \to \mathbb{R}^{d-1}$, $\Delta z = z(d) - z(d-1) = 1$. *The holographic bit is preserved:* $\Delta(\Delta S) = 0$.

Proof. $\Delta z = (d+1) - d = 1$. $\Delta D_V = \Delta D_A = \ln(d+1) - \ln d$. $\Delta(\Delta S) = 0$. □

5 The new identity and the dark matter fraction

Physical meaning of $z_{\text{pack}} = 12$ in the vacuum context. The classical kissing number $z_{\text{pack}}(\mathbb{R}^3) = 12$ (Schütte & van der Waerden [3]) is a purely mathematical theorem about the maximal number of non-overlapping unit spheres that can touch a central unit sphere; by itself, it carries no physical content specific to a quantum vacuum. Its appearance in Proposition 5.1 and throughout Section 5 as the quantity subtracted from $z_{\text{fund}} = 4$ to define the "geometric frustration" of the vacuum requires an independent physical justification for *why* this particular classical bound is the relevant one to compare against z_{fund}, rather than some other geometric invariant of $\mathbb{R}^3$.

This justification is given in TEG v8 (Section 3.1) and is reproduced here for self-containedness, since the present derivation chain depends on it: $z_{\text{pack}} = 12$ maximises *mass density* (it is the densest possible sphere-packing coordination, the classical kissing-number problem), whereas $z_{\text{fund}} = 4$ maximises *information density per boundary node* (Proposition 3.1, Theorem 4.1). The vacuum network is conjectured to carry quantum information rather than packed mass; under this reading, $z_{\text{pack}} - z_{\text{fund}}$ counts the links per node that are geometrically available to dense matter but left unoccupied by the information-carrying vacuum structure — the quantity TEG calls geometric frustration. Without this distinction, $z_{\text{pack}} - z_{\text{fund}} = 8$ (Proposition 5.1) is simply an arithmetic difference between two independently true theorems (the kissing number and the minimal-simplex facet count), with no a priori reason to treat their difference as physically meaningful. The mass-vs-information distinction is what licenses calling this difference "frustration" rather than an arbitrary subtraction, and it is an assumption inherited from TEG v8's Hypothesis H0, not a result proved within the present (vH2) derivation chain.

We flag this explicitly: the chain from Layer 0 to Ω_{DM} (Section 8) is algebraically rigorous given $z_{\text{pack}} = 12$ and $z_{\text{fund}} = 4$ as inputs, but its *physical* interpretation as "frustration" — and therefore its claim to predict a cosmological dark-matter fraction rather than merely compute a number from two true theorems — rests on the matter/information distinction of TEG v8, which is itself stated there as a physical hypothesis (H0) rather than a derived result. This assumption is listed explicitly as Input I4 in the complete logical chain of Section 8.

5.1 A new identity connecting three independent results

Proposition 5.1 (Kissing–simplex–bits identity)**.** *In $\mathbb{R}^3$:*

$$z_{\text{pack}}(\mathbb{R}^3) - z(3) \;=\; 12 - 4 \;=\; 8 \;=\; 2^{N_{\text{bits}}(3)}. \tag{2}$$

This identity is unique to $d = 3$: it fails for all other dimensions where z_{pack} is known.

Proof. Direct computation: $12 - 4 = 8 = 2^3$. For $d = 2$: $z_{\text{pack}} - z(2) = 3 \neq 2^{2.585}$. For $d = 4$: $z_{\text{pack}} - z(4) = 19 \neq 2^{3.322}$. The identity requires $N_{\text{bits}} \in \mathbb{Z}$, which by Theorem 4.4 holds only for $d = 3$. □

The identity connects three independent domains:

- $z_{\text{pack}}(\mathbb{R}^3) = 12$: the kissing number [3];
- $z(3) = 4$: the minimal simplex (Theorem 4.1);
- $N_{\text{bits}}(3) = 3$: the holographic bit count (Theorem 4.4).

5.2 Explanation of the identity: a corollary of Theorem 4.4

The kissing–simplex–bits identity is not an independent result. We show it follows from Theorem 4.4 via a new factorisation of the kissing number.

Proposition 5.2 (Kissing number factorisation). *In $\mathbb{R}^d$:*

$$z_{\text{pack}}(\mathbb{R}^d) \;=\; N_{\text{bits}}(d) \cdot z(d) \tag{3}$$

if and only if $d = 3$.

Proof. (d = 3, direct): $N_{\text{bits}}(3) \cdot z(3) = 3 \times 4 = 12 = z_{\text{pack}}(\mathbb{R}^3)$. ✓

(Uniqueness): The right-hand side of eq. (3) is an integer if and only if $N_{\text{bits}}(d) \in \mathbb{Z}$. By Theorem 4.4, $N_{\text{bits}}(d) \in \mathbb{Z}$ holds only for $d = 3$. For all $d \neq 3$, $N_{\text{bits}}(d) \notin \mathbb{Z}$, so $N_{\text{bits}}(d) \cdot z(d) \notin \mathbb{Z}$ and cannot equal the integer $z_{\text{pack}}(\mathbb{R}^d)$. □

Remark 5.3 (Constructive proof via the cuboctahedron). The proof of the "if" direction in Proposition 5.2 can be made fully constructive. Define the contact set of the FCC lattice:

$$\mathcal{C} \;:=\; \big\{\, \mathbf{v} \in \mathbb{Z}^3 : |\mathbf{v}|^2 = 2,\ \text{exactly one coordinate of } \mathbf{v} \text{ is zero} \,\big\}.$$

Lemma (cuboctahedron decomposition).

(i) $|\mathcal{C}| = 12 = z_{\text{pack}}(\mathbb{R}^3)$.

(ii) $\mathcal{C}$ decomposes into exactly three disjoint square sections: $\mathcal{S}_k = \{\mathbf{v} \in \mathcal{C} : v_k = 0\}$, $k = 1, 2, 3$.

(iii) $|\mathcal{S}_k| = 4 = z(3)$ for each k.

Proof. (i) Choose which coordinate is zero: $d = 3$ choices. Choose the signs of the remaining two coordinates: $2^2 = 4$ choices. Total: $3 \times 4 = 12$. (ii) Each $\mathbf{v} \in \mathcal{C}$ has *exactly* one zero coordinate, so the sets $\mathcal{S}_k$ are pairwise disjoint by definition, and their union is $\mathcal{C}$. (iii) Fixing k, the remaining two entries range over $\{(\pm1, \pm1)\}$: exactly $2^2 = 4$ vectors. □

Corollary. $z_{\text{pack}}(\mathbb{R}^3) = 3 \times 4 = N_{\text{bits}}(3) \times z(3)$.

Why only $d = 3$? In step (iii), fixing one zero coordinate leaves $d - 1 = 2$ free coordinates with signs ±1, giving 2^{d-1} vectors per section. The factorisation $z_{\text{pack}} = d \cdot 2^{d-1}$ holds for the FCC analogue only when $2^{d-1} = z(d) = d + 1$, i.e. exactly when $d = 3$ (Theorem 4.4). For $d = 8$ the optimal lattice is E_8, which lacks the coordinate-aligned cubic symmetry of FCC, so the decomposition into d sections of 2^{d-1} vertices each does not extend.

Corollary 5.4 (Identity as corollary of Prop. 5.2 and Thm. 4.4). *The kissing–simplex–bits identity follows algebraically from Proposition 5.2 and Theorem 4.4:*

$$\begin{aligned}
z_{\text{pack}}(\mathbb{R}^3) - z(3) &= N_{\text{bits}} \cdot z(3) - z(3) && \textit{(Prop. 5.2)}\\
&= z(3)\,(N_{\text{bits}} - 1) && \textit{(factoring)}\\
&= 4 \times 2 = 8 && (z(3) = 4,\ N_{\text{bits}} = 3)\\
&= 2^3 = 2^{N_{\text{bits}}}. && \textit{(Thm. 4.4: } d + 1 = 2^{d-1} \textit{ at } d = 3)
\end{aligned}$$

The last step uses $z(3)(N_{\text{bits}} - 1) = 2^{N_{\text{bits}}}$, which is equivalent to $d + 1 = 2^{d-1}$ at $d = 3$—precisely the condition of Theorem 4.4. The identity therefore has a single number-theoretic origin: $d + 1 = 2^{d-1}$ holds if and only if $d = 3$.

The kissing–simplex–bits identity is therefore a consequence of the single number-theoretic fact $d + 1 = 2^{d-1}$ at $d = 3$, the same fact that underlies all four instances of the 3/2 ratio (Section 6).

5.3 Geometric frustration and dark matter

The $z_{\text{pack}} - z_{\text{fund}} = 8$ links per node that are geometrically available but not occupied by the minimal simplex constitute the *geometric frustration* of the vacuum.

Theorem 5.5 ($\mathbb{Z}_2$ symmetry of the S^3 sigma-model)**.** *The vacuum potential $V(a, \rho)$ of the non-linear sigma-model on S^3 satisfies $V(-a, \rho) = V(a, \rho)$ for all $a \in (-1, 1)$ and all ρ. Consequently, the partition function of a frustrated link satisfies $Z_{\text{frust}} = 2Z_+$ exactly, where $Z_+ = \int_0^{\pi/2} \sin^2\theta \, e^{-V(\cos\theta)} \, d\theta$ is the contribution of the $a > 0$ hemisphere.*

Proof. Every term of $V(a, \rho)$ depends only on a^2 or $(1 - a^2)$:

$$\begin{aligned} & D_A(1 - a^2) && \text{(function of } a^2\text{)}, \\ -\tfrac{3}{2}S(a) \; = \; & \tfrac{3}{2}\big[(1 - a^2)\ln(1 - a^2) + a^2 \ln a^2\big] && \text{(function of } a^2\text{)}, \\ & -\tfrac{\pi}{4}\ln(1 - a^2) && \text{(function of } a^2\text{)}, \\ & \sigma_{\text{eff}}(\rho/\rho_c)\,\partial\, a^2 && \text{(function of } a^2\text{)}. \end{aligned}$$

Therefore $V(-a, \rho) = V(a, \rho)$ exactly. The measure $\sin^2\theta \, d\theta$ on S^3 satisfies $\sin^2(\pi - \theta) = \sin^2\theta$, so it is also invariant under $\theta \mapsto \pi - \theta$ (i.e. $a \mapsto -a$). Hence

$$Z_- \; := \; \int_{\pi/2}^{\pi} \sin^2\theta \, e^{-V(\cos\theta)} \, d\theta \; = \; Z_+,$$

and $Z_{\text{frust}} = Z_+ + Z_- = 2Z_+$. □

Remark 5.6 (Physical interpretation). The two hemisferios $a > 0$ and $a < 0$ correspond to the two orientational states of a frustrated link relative to the S^3 equator: the scalar component a points either toward or away from the projected $\mathbb{R}^3$ vacuum. This is the interior/exterior duality of Section 2, now derived rather than postulated. The factor 2 is verified numerically to machine precision: $Z_{\text{total}}/Z_+ = 2.0000000000$ (Appendix A).

Using Theorem 5.5, the absolute frustration is:

$$F(\mathbb{R}^3) \; = \; \frac{D_V}{D_A} \cdot \ln\!\left(\frac{z_{\text{pack}}}{z_{\text{pack}} - z_{\text{fund}}}\right) \cdot 2 \; = \; \frac{3}{2} \cdot \ln\frac{12}{8} \cdot 2 \; = \; 2\ln\frac{3}{2}. \tag{4}$$

The factor 2 is the exact consequence of the $\mathbb{Z}_2$ symmetry $q \mapsto \sigma(q) = -a + \mathbf{v}$ of S^3.

Remark 5.7 (Identification of the factor 2 in D_V). The factor 2 in $D_V = \ln(2z_{\text{fund}})$ is introduced in Definition **??** as a postulate, motivated by the interior/exterior orientational duality of the facets of the minimal simplex. Theorem **??** derives it independently as an exact consequence of the $\mathbb{Z}_2$ symmetry $a \mapsto -a$ of the S^3 sigma-model, where $Z_{\text{frust}} = 2Z_+$ exactly (numerically verified: $Z_{\text{total}}/Z_+ = 2.0000000000$, see Appendix A).

The physical identification between the combinatorial property ("each facet has an interior and an exterior side") and the analytic property ("$V(a) = V(-a)$") is reasonable but does not yet constitute a rigorous mathematical proof that these are the same object, rather than two numerically equal quantities. Establishing this formal equivalence is part of Open Problem 2.

5.4 The dark matter fraction

Proposition 5.8 (Equipartition of the frustration). *Let the quaternionic vacuum be described by the field $q \in S^3 \subset \mathbb{H}$ with S_3 isotropy symmetry (Axiom 2.1). The S_3-invariant maximum-entropy state on the Hilbert space $(\mathbb{C}^2)^{\otimes N_{\text{bits}}}$, with $N_{\text{bits}} = 3$, is*

$$\rho_{\text{total}} = \frac{I_8}{8}.$$

The partial trace over any two subsystems satisfies

$$S(\rho_x) = \ln 2, \qquad \frac{S(\rho_x)}{S(\rho_{\text{total}})} = \frac{\ln 2}{3\ln 2} = \frac{1}{N_{\text{bits}}},$$

and the geometric frustration F is equipartitioned among the $N_{\text{bits}} = 3$ spatial degrees of freedom:

$$\Omega_{\text{DM}} = \frac{F(\mathbb{R}^3)}{N_{\text{bits}}}.$$

Proof. The quaternionic vacuum carries $N_{\text{bits}} = 3$ spatial qubits (Theorem **??**). The associated Hilbert space is $\mathcal{H} = (\mathbb{C}^2)^{\otimes 3}$, of dimension 8.

Step 1 – Uniqueness of the S_3 maximum-entropy state. The isotropy symmetry of Axiom 2.1 requires ρ_{total} to commute with every permutation of spatial axes:

$$[\rho_{\text{total}}, P_{ij}] = 0 \quad \forall P_{ij} \in S_3.$$

Under the SU(2) decomposition $(\frac{1}{2})^{\otimes 3} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}$, every S_3-invariant state is block-diagonal. The entropy $S = -\text{Tr}(\rho \ln \rho)$ is maximised uniquely when all blocks are maximally mixed, giving $\rho_{\text{total}} = I_8/8$.

Step 2 – Partial entropy. Tracing over any pair of subsystems gives $\rho_x = \text{Tr}_{yz}(I_8/8) = I_2/2$, with von Neumann entropy

$$S(\rho_x) = -\text{Tr}\big(\tfrac{I_2}{2} \ln \tfrac{I_2}{2}\big) = \ln 2.$$

This coincides exactly with the holographic bit $\Delta S = D_V - D_A = \ln 2$ (Theorem **??**).

Step 3 – Equipartitioned fraction. The fraction of frustration per degree of freedom is

$$\frac{S(\rho_x)}{S(\rho_{\text{total}})} = \frac{\ln 2}{3\ln 2} = \frac{1}{3} = \frac{1}{N_{\text{bits}}},$$

an exact integer guaranteed only at $d = 3$ by Theorem **??**. The total frustration $F(\mathbb{R}^3)$ is therefore distributed as $\Omega_{\text{DM}} = F/N_{\text{bits}}$.

Dependency note. The complete proof of Steps 1–2, including the explicit numerical diagonalization of the Casimir operator and verification of the tetrahedral closure condition, is given in [1], Appendix E, Theorems 2 and 3. The present block extracts the three logical steps essential for vH2 to be self-contained; the numerical verification is reproduced in Appendix A (see P8 below). □

Theorem 5.9 (Dark matter fraction). *Under the equipartition theorem of TEG v8, the cosmological dark matter fraction is:*

$$\boxed{\Omega_{\text{DM}} = \frac{F(\mathbb{R}^3)}{N_{\text{bits}}} = \frac{2\ln(3/2)}{3} \approx 0.2703.} \tag{5}$$

Proof. $F(\mathbb{R}^3) = 2\ln(3/2)$ is established by eq. (4) and Theorem 5.5 (no conjecture required). The frustration is distributed equipartitely among $N_{\text{bits}} = 3$ spatial degrees of freedom (Theorem 4.4 and the equipartition theorem of Ref. [1]). Each bit carries equal gravitational weight, giving $\Omega_{\text{DM}} = F/N_{\text{bits}} = 2\ln(3/2)/3$. □

Table 5: Comparison of the algebraic prediction with observations.

Source	Ω_{DM}	Discrepancy
TEG v4 [eq. (5)]	$2\ln(3/2)/3 = 0.27031$	—
SH0ES [10]	0.270 ± 0.010	0.1%
Planck 2018 [5]	0.2589 ± 0.0057	4.4%

Remark 5.10 (Implied cosmological constant). With $\Omega_b = 0.049$ and the flat-universe constraint:

$$\Omega_\Lambda = 1 - \Omega_{\text{DM}} - \Omega_b = 1 - \frac{2\ln(3/2)}{3} - 0.049 \approx 0.681$$

(Planck 2018: $\Omega_\Lambda = 0.691$, discrepancy 1.5%).

6 Four instances of the 3/2 ratio

All four occurrences of 3/2 in TEG follow from $z(3) = d + 1 = 4 = 2^{d-1}$, the unique solution of $d + 1 = 2^{d-1}$ at $d = 3$.

Table 6: Four instances of 3/2 and their common geometric origin.

Context	Expression	Origin
Entropy ratio	$D_V/D_A = \ln 8/\ln 4 = 3/2$	$z(3) = 4 = 2^2$
Sphere packing	$z_{\text{pack}}/(z_{\text{pack}} - z_{\text{fund}}) = 12/8 = 3/2$	$z_{\text{pack}} - z_{\text{fund}} = 2^{N_{\text{bits}}}$
BH temperature	$T_{\text{TEG}}/T_H = 3/2$	$S_{\text{TEG}} = \frac{2}{3}S_{\text{BH}}$
Dark matter	$\Omega_{\text{DM}} = \frac{2}{3}\ln\frac{3}{2}$	Frustration$/N_{\text{bits}}$

7 Scalar field dynamics and the Newtonian limit

7.1 Non-linear sigma-model on S^3

Since $|q| = 1$, the vacuum lives on S^3. The scalar component a parameterises the polar angle via $a = \cos\theta$. The kinetic term is:

$$\mathcal{L}_{\text{kin}} = \frac{(\partial_\mu a)(\partial^\mu a)}{2(1 - a^2)}.$$

7.2 The density-dependent potential

$$V(a, \rho) = D_A(1 - a^2) - \frac{3}{2}S(a) - \frac{\pi}{4}\ln(1 - a^2) + \sigma_{\text{eff}}\frac{\rho}{\rho_c}\,\partial\, a^2, \tag{6}$$

where $S(a) = -(1-a^2)\ln(1-a^2) - a^2 \ln a^2$, $\partial = 3 - \ln 8 \approx 0.9206$, no free parameters. The functional form is motivated by the quaternion geometry but not yet derived from a variational principle (Open Problem 3).

Renormalised potential for the covariant formulation. Appendix F shows that, within the covariant action S_{TEG}, the field equations require the renormalised potential

$$\tilde{V}(a,\rho) \equiv V(a,\rho) - D_A, \tag{7}$$

where $D_A = \ln 4$ is the surface entropy of the minimal simplex (Definition 4.2). The boundary condition $\tilde{V}(0) = 0$ ensures that the high-density vacuum ($a \to 0$) contributes zero cosmological energy, recovering GR in the Solar System (Appendix F.5).

Two properties are preserved under $V \to \tilde{V}$: (i) all rotation-curve results are unchanged, since they depend on $\partial\tilde{V}/\partial a = \partial V/\partial a$; (ii) the Z_2 symmetry $a \mapsto -a$ (Theorem 5.5) is preserved. The value $V(0) = D_A$ is geometrically meaningful (it equals the tetrahedral surface entropy), but its role as a cosmological constant is removed by the physical boundary condition (7).

Note on the coefficients of Eq. (6). Of the four terms in $V(a,\rho)$, three have an identified geometric origin and one does not:

- $D_A(1-a^2)$: the coefficient $D_A = \ln 4$ is the surface entropy of Definition 4.2.
- $-\frac{3}{2}S(a)$: the coefficient $3/2$ is *not* an independent input. It is algebraically identical to $D_V/D_A = \ln 8/\ln 4 = 3/2$ (Theorem 4.5), i.e. the same ratio appearing in Table 5 as the entropy-ratio instance of the universal $3/2$. We record this explicitly here: the weight of the von Neumann entropy term in $V(a,\rho)$ is the structural ratio D_V/D_A, not an independently chosen number. This does not yet derive the term from a variational principle (Open Problem 5), but it removes one degree of freedom that could otherwise be mistaken for an unconstrained fit parameter.
- $-\frac{\pi}{4}\ln(1-a^2)$: no geometric origin specific to the tetrahedral axiom has been identified for the coefficient $\pi/4$. The nearest TEG-derived constant is $\ln 2 = 0.6931$, which differs from $\pi/4 = 0.7854$ by 13.3% — too large to be considered a match given the machine-precision identities found elsewhere in TEG (cf. Proposition 2.2, Theorem 4.3). We record $\pi/4$ honestly as an *undetermined coefficient*: it plausibly reflects a general normalization constant of S^3 geometry (such factors are common in integrals over the 3-sphere) rather than a quantity tied specifically to $z_{\mathrm{fund}} = 4$. Its derivation, or replacement by an explicitly derived constant, remains part of Open Problem 5.
- $\sigma_{\mathrm{eff}}(\rho/\rho_c)^{\partial} a^2$: σ_{eff} and ∂ are both independently derived (Appendix E; Section 6.3 of TEG v8).

In summary, two of the four terms in $V(a,\rho)$ rest on fully derived coefficients, one rests on a coefficient that is itself an exact algebraic identity among already-derived quantities ($3/2 = D_V/D_A$), and one ($\pi/4$) remains an undetermined input. This sharpens, rather than resolves, Open Problem 5: the remaining gap is now isolated to a single coefficient instead of the whole potential.

7.3 The Newtonian transition

Proposition 7.1 (Newtonian transition threshold)**.** *The Newtonian regime* $\Phi \to 1$ *is reached when*

$$\frac{\rho}{\rho_c} \gtrsim \left(\frac{D_A}{\sigma_{\text{eff}}}\right)^{1/\partial} \approx 15.9, \tag{8}$$

derived analytically from $D_A = \ln 4$, $\sigma_{\text{eff}} = 0.1088$, $\partial = 3 - \ln 8$ *with no free parameters.*

Table 7: Equilibrium scalar field and amplification factor vs. density.

ρ/ρ_c	a^*	$\Phi^* = (1 - a^{*2})^{-1/2}$
0	0.6925	$1.386 \approx \ln 4$
1.0	0.6836	1.370
10.0	0.6114	1.264
15.9	0.5630	1.210
100	0.0981	$1.005 \approx 1$

The equation of motion $\Box a + (1 - a^2)\partial V/\partial a = 0$ is structurally identical to the chameleon equation [7]: $m^2_{\text{eff}}(\rho)$ grows with ρ, driving $a \to 0$ in dense environments and decoupling the scalar from Solar System tests.

The vacuum temperature ratio $T_{\text{vac}} = D_V/D_A = 3/2$ is identical to $T_{\text{TEG}}/T_H = 3/2$ derived independently in TEG v8: a non-trivial internal consistency check.

Inherited caveats from TEG v8 on the density-dependent mechanism. Proposition 7.1 establishes the threshold $\rho/\rho_c \gtrsim 15.9$ analytically, with no free parameters, and the limit $\Phi \to 1$ as $\rho/\rho_c \to \infty$ is correct and unconditional. However, the quantitative performance of the underlying mechanism $\Phi(\rho)$ (Eq. 5) on real rotation-curve data was tested extensively in TEG v8 (Section 10.3, Appendices I–J), and those results carry directly over to this version since the mechanism is unchanged. We summarize them here rather than leaving them to be located in a companion document.

- On 171 SPARC galaxies, $\Phi(\rho)$ with a universal ρ_c *improves* massive galaxies ($\log M_b > 10.5$, $N = 57$) by 0.033 dex on average, but *degrades* low-mass galaxies ($\log M_b < 10.5$, $N = 114$) systematically, and worsens the global RMSE by +0.0054 dex relative to the baseline without $\Phi(\rho)$ (TEG v8, Table 17).

- The transition is sharp at $\log M_b \approx 10.5$, with no current explanation within the tetrahedral axiom (TEG v8, Open Problem 8).

- A second zero-parameter prediction derived from the same axiom, the geometric scaling exponent $\alpha = 1 - 3/\ln 8 = -0.4427$ for the critical density $\rho_c \propto M_b^\alpha$, was tested on the same data and *falsified*: it degrades all six mass bins without exception (TEG v8, Appendix I.3).

Honest status. The Newtonian-limit recovery $\Phi \to 1$ established by Proposition 7.1 is a correct asymptotic statement, independent of these caveats. What is not yet established is a density-dependent mechanism that improves rotation-curve fits *uniformly* across the full SPARC mass range; the present form of $\Phi(\rho)$ helps massive galaxies and hurts low-mass ones, and the natural geometric candidate for the transition scale has been tested and ruled out. Resolving this — connected to Open Problems 2, 3, and 8 of TEG v8 — remains required before the chameleon mechanism can be considered a complete solution to the Newtonian-limit problem, rather than a partial one.

8 Complete logical chain

The derivation proceeds in three layers. We distinguish *physical inputs* (axioms and classical theorems taken as given) from *derived results* (proved from those inputs alone).

Layer 0 — Physical inputs (not derived within TEG).

[label=**I8.**]

1. **Quaternion Vacuum Axiom** (Axiom 2.1): the quantum vacuum is a unit-quaternion field on $\mathbb{H} \cong S^3$; observed physics is the projection $\pi : \mathbb{H} \to \mathbb{R}^3$.

2. **5-cell selection** (Proposition 3.1): is proved by direct computation within TEG but is not derived from a dynamical principle (Open Problem 2 addresses this). The two projections π_{alg} and π_{geo} are shown to be physically equivalent in Theorem 3.4.

3. **Kissing number** (classical, [3]): $z_{\text{pack}}(\mathbb{R}^3) = 12$.

4. **Matter/information distinction (Hypothesis H0, inherited from [1]).** The identification of $z_{\text{pack}} - z_{\text{fund}} = 8$ as *geometric frustration* — rather than an arithmetic difference between two independently true theorems — requires the physical hypothesis that the quantum vacuum carries information rather than packed mass. This is Hypothesis H0 of [1]: $z_{\text{pack}} = 12$ bounds mass-density coordination, while $z_{\text{fund}} = 4$ maximises information-density per boundary node (Proposition 3.1); their difference counts links available to dense matter but left unoccupied by the information-carrying vacuum structure. Without this distinction, $z_{\text{pack}} - z_{\text{fund}} = 8$ is not a cosmological quantity. This hypothesis is stated in [1] as a physical assumption, not a derived result, and is inherited here with the same status.

Layer 1 — Structural theorems (proved from inputs).

$$\underbrace{z(d) = d + 1}_{\text{Thm. 4.1}} \xrightarrow{d=3} z_{\text{fund}} = 4 \xrightarrow{\text{Thm. 4.4}} N_{\text{bits}} = 3 \xrightarrow{\text{Thm. 4.5}} D_V/D_A = 3/2$$

Layer 2 — Dark matter fraction (proved from Layers 0–1).

$$\underbrace{z_{\text{pack}} = 12}_{\textbf{I3}} \implies \underbrace{z_{\text{pack}} = N_{\text{bits}} \cdot z(3)}_{\text{Prop. 5.2}} \implies \underbrace{z_{\text{pack}} - z_{\text{fund}} = 2^{N_{\text{bits}}}}_{\text{Cor. 5.4 + Thm. 4.4}} \xrightarrow{\text{Thm. 5.5}} \mathcal{F} = 2\ln\tfrac{3}{2} \xrightarrow{\div N_{\text{bits}}} \Omega_{\text{DM}}$$

Remark 8.1 (Scope of the logical scheme). Every arrow in Layer 2 is a proved theorem or corollary (Theorems 5.1–5.5). However, the interpretation of $z_{\text{pack}} - z_{\text{fund}}$ as the vacuum's "geometric frustration" — and therefore the reading of Ω_{DM} as a cosmological fraction rather than a number computed from two independently true theorems — rests on the matter/information distinction of Input I4, which is a physical hypothesis (H0 of [1]), not a result proved within vH2's own algebraic chain.

The claim of TEG vH3 is therefore precisely: *given* I1–I4, $\Omega_{\mathrm{DM}} = 2\ln(3/2)/3$ follows with no additional free parameters or conjectures. The first-principles justification of I4 is part of Open Problem 2.

Remark on scope. Input **I1** (the quaternion vacuum) is a physical axiom whose ultimate justification lies outside TEG. Input **I2** (5-cell selection) is proved by direct computation within TEG but is not derived from a dynamical principle (Open Problem 2 addresses this). Input **I3** (kissing number) is a classical mathematical theorem. The claim of TEG v4 is therefore: *given **I1–I3**, $\Omega_{\mathrm{DM}} = 2\ln(3/2)/3$ follows with no additional free parameters or conjectures.*

9 Predictions and falsification

1. **Dark matter fraction.** $\Omega_{\mathrm{DM}} = 2\ln(3/2)/3 \approx 0.2703$. Falsified if the measured value departs by more than 2σ after resolving the Hubble tension.

2. **No dark matter particle.** Dark matter is geometric. XENON, LUX, PandaX, XENONnT yield null results at any sensitivity.

3. **Newtonian transition at $\rho/\rho_c \approx 15.9$,** with no free parameters.

4. **Hubble constant (Conjecture 3).** $H_0^{\mathrm{TEG}} = H_{\mathrm{CMB}}/\sqrt{\partial} = 70.25\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, where $\partial = 3 - \ln 8 = D_{\mathrm{spatial}} - D_{\mathrm{eff}}$ is the fractal codimension (Appendix F, Eq. 39).

   This is **Conjecture 3**, not a derived result. Within the covariant formulation (Appendix F), the field equations are standard Einstein equations, which in FRW give $H^2 = (8\pi G/3)\rho$ without the factor $1/\partial$. The factor $1/\partial$ requires additional structure — a non-minimal coupling of a to R, or contributions from $\tilde{V}(a^*, \rho_{\mathrm{CMB}})$ — not present in the current action.

   An alternative structural estimate (Route B, Appendix F.7.2) gives $H_0^B = 61.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, based on the fractional operator in FRW. The discrepancy $|H_0^A - H_0^B| = 8.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ constitutes the primary open sub-problem of the covariant extension. Conjecture 3 retains Route A for three reasons documented in Appendix F.7.3: geometric transparency, fewer approximations, and better observational concordance (0.3σ of CCHP 2025, 1.5σ of DESI 2024). Falsified if the measured H_0 converges to $< 65\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ with $< 1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ uncertainty.

5. **Four-fold** 3/2 **ratio.** Any simultaneous violation of $D_V/D_A = 3/2$, $T_{\mathrm{TEG}}/T_H = 3/2$, or $\Omega_{\mathrm{DM}} = \frac{2}{3}\ln\frac{3}{2}$ falsifies the geometric origin of 3/2.

6. **Lensing–dynamics decoupling.** $M_{\mathrm{lens}}/M_{\mathrm{dyn}} \neq 1$ in low-density environments.

10 Open problems

Open Problem 1 (Baryon fraction from spectral entropy of K_5). **Derivation of α^* from TEG.** The entropic coupling α^* is not a free parameter: it is the holographic bit density of the projected vacuum, defined as the ratio of the information lost per node to the square of the surface entropy:

$$\alpha^* = \frac{\Delta S}{D_A^2} = \frac{\ln 2}{(\ln 4)^2} = \frac{1}{4\ln 2} \approx 0.3607. \tag{9}$$

This is the TEG analogue of the Bekenstein–Hawking area density $1/(4G)$, with $\ln 2$ playing the role of G. It follows from two already-proved results: $\Delta S = \ln 2$ (Theorem 4.3) and $D_A = \ln 4$ (Theorem 4.1). No external input is needed.

Spectral entropy of K_5. Let $L_{K_5}(\alpha^*)$ be the weighted Laplacian of the complete graph K_5 (the 1-skeleton of the 5-cell) with the TEG causal partition $(1, 3, 1)$: vertex $B = \{a\}$ (scalar), vertices $C = \{v_1, v_2, v_3\}$ (spatial), vertex $E = \{e\}$ (temporal), with weight α^* on edges incident to B and weight 1 on all other edges. The non-zero eigenvalues are:

$$\lambda_1 = 5\alpha^*, \qquad \lambda_2 = \lambda_3 = \lambda_4 = \alpha^* + 4, \tag{10}$$

with spectral Shannon entropy $S_{K_5} = -p_1 \ln p_1 - 3p_{234} \ln p_{234}$, where $p_1 = 5\alpha^*/(8\alpha^* + 12)$ and $p_{234} = (\alpha^* + 4)/(8\alpha^* + 12)$.

Baryon fraction candidate. The spectral entropy deficit of K_5 relative to the projected tetrahedron is:

$$\boxed{\Omega_b^{(K_5)} \;=\; D_A - S_{K_5} \;=\; \ln 4 - S_{K_5} \;\approx\; 0.0516,} \tag{11}$$

within 5.2% of the observed $\Omega_b = 0.049$, derived entirely from TEG constants with no free parameters.

Physical interpretation. $D_A = \ln 4$ is the surface entropy of the minimal simplex. S_{K_5} is the information content of the spectral distribution of K_5 under the causal weighting. The deficit $D_A - S_{K_5}$ measures the entropy the 5-cell owes to the projected tetrahedron: this residual is baryonic matter. The split $\Omega_{\mathrm{DM}} \sim D_V$ (bulk frustration) vs. $\Omega_b \sim D_A$ (surface deficit) is consistent with $D_A/D_V = 2/3$ throughout TEG.

Flat-universe check. $\Omega_{\mathrm{DM}} + \Omega_b^{(K_5)} + \Omega_\Lambda \;=\; 0.2703 + 0.0516 + 0.6781 \;=\; 1.000$, with $\Omega_\Lambda \approx 0.678$ versus Planck 0.691 (error 1.9%).

What remains open. The residual 5.2% error on Ω_b likely reflects the causal structure of the partition: the $(1, 3, 1)$ partition treats B (scalar, past) and E (temporal, future) symmetrically, giving $\eta = 0$ for the baryon asymmetry. Breaking this symmetry requires identifying the correct causal asymmetry in the 5-cell that encodes baryogenesis. Note that Blatière [9] independently derives $\alpha^* = 1/(4 \ln 2)$ from a different causal partition $(2, 1, 2)$ of the same 5-cell, obtaining $\eta \approx 5.93 \times 10^{-10}$ (Planck: 6.10×10^{-10}). The two frameworks share the polytope and the coupling; the difference is the causal partition, suggesting a joint derivation of both Ω_b and η from a single TEG causal structure.

Open Problem 2 (Derivation of the vacuum potential from first principles). Theorem 5.5 derives the factor 2 in $F = 2\ln(3/2)$ from the $\mathbb{Z}_2$ symmetry $a \mapsto -a$ of the S^3 sigma-model, closing the only previously conjectural step in the derivation of Ω_{DM}. What remains is to derive the *full* potential $V(a, \rho)$ (eq. (6)) from a single variational principle in the S^3 sigma-model, rather than motivating each term separately from the quaternion geometry. This would unify the $\mathbb{Z}_2$ result with the chameleon mechanism and the Newtonian transition in a single action.

Open Problem 3. (Covariant formulation — partial resolution). A covariant action for TEG has been constructed (Appendix F, Eq. 38) using the multi-fractional spacetime formalism of Calcagni [24]. The main results are:

1. The single free parameter of Calcagni's family is fixed exactly by the tetrahedral axiom: $\alpha_{\mathrm{TEG}} = \ln 8/4 = (3/4) \ln 2$.

2. The field equations are standard Einstein equations sourced by $T^{(\mathrm{bar})} + T^{(a)}$; the multi-fractional measure cancels from the gravitational sector.

3. GR is recovered exactly in the high-density limit with the renormalised potential $\tilde{V} = V - D_A$ (boundary condition $\tilde{V}(0) = 0$).
4. The holographic codimension $\partial = 3 - \ln 8$ is identified as $\partial = D_{\text{spatial}} - D_{\text{eff}}$, giving it a first-principles geometric interpretation.
5. The fractal correction to the scalar EOM, Δ_{frac}, exists but is suppressed by $(H_0/H_{\text{Pl}})^2 \approx 10^{-122}$ in the IR. All main-text results are unchanged.

Three sub-problems remain open:

(3A) The Friedmann equation has not been derived from S_{TEG} in FRW geometry. Two conjectural routes give $H_0 = 70.25$ (Route A) and $H_0 = 61.8$ km/s/Mpc (Route B); reconciling them requires the FRW heat kernel at $\alpha = \ln 8/4$.

(3B) The vacuum mass profile $M_{\text{vac}}(r)$ must be derived from $\tilde{V}(a, \rho)$ through the full scalar dynamics.

(3C) The boundary condition $\tilde{V}(0) = 0$ resolves the vacuum energy problem but lacks derivation from a symmetry principle or renormalisation group argument.

Sub-problems (3A)–(3C) are identified as the primary targets of the next TEG publication.

Open Problem 4 (Planck vs. SH0ES discrepancy). Joint derivation of H_0 and Ω_{DM} from the same geometric axiom to resolve the 4.4% gap.

Open Problem 5 (Dimension-independence of the factorisation). Remark 5.3 gives a complete constructive proof that $z_{\text{pack}}(\mathbb{R}^3) = N_{\text{bits}}(3) \cdot z(3)$ via the cuboctahedron decomposition of the FCC contact set, and identifies why it fails for $d = 8$ (the E_8 lattice lacks coordinate-aligned cubic symmetry). What remains is a unified theorem characterising *all* dimensions in which the contact set of the optimal packing decomposes as $z_{\text{pack}}(\mathbb{R}^d) = d \cdot 2^{d-1}$, and proving that $d = 3$ is the unique dimension where this coincides with $N_{\text{bits}}(d) \cdot z(d)$.

11 Conclusion

Starting from the quaternion vacuum axiom and the Minimal Simplex Principle, together with two classical theorems (5-cell entropy maximisation in S^3; $z_{\text{pack}}(\mathbb{R}^3) = 12$ [3]), we have derived:

1. $z_{\text{fund}} = 4$ is a theorem, not a hypothesis: the projection of the 5-cell (Section 3).
2. $N_{\text{bits}} = 3$ is exact and unique to $d = 3$ (Theorem 4.4).
3. $D_V/D_A = 3/2$ is exact and unique to $d = 3$ (Theorem 4.5).
4. Newtonian limit resolved via chameleon mechanism, threshold $\rho/\rho_c \approx 15.9$, no free parameters (Proposition 7.1).
5. New factorisation: $z_{\text{pack}}(\mathbb{R}^3) = N_{\text{bits}} \cdot z(3) = 3 \times 4$, unique to $d = 3$ (Proposition 5.2), proved constructively via the cuboctahedron decomposition of the FCC contact set (Remark 5.3).

6. The kissing–simplex–bits identity $z_{\text{pack}} - z(3) = 2^{N_{\text{bits}}}$ follows as a corollary (Corollary 5.4).

7. The factor 2 in $F = 2\ln(3/2)$ is the exact consequence of the $\mathbb{Z}_2$ symmetry $a \mapsto -a$ of the S^3 sigma-model (Theorem 5.5).

8. $\Omega_{\text{DM}} = 2\ln(3/2)/3 \approx 0.2703$, within 0.1% of local measurements, **with no conjectures** (Theorem 5.9).

9. The entropic coupling $\alpha^* = \Delta S/D_A^2 = 1/(4\ln 2)$ is derived from TEG constants with no external input (eq. (9)).

10. Baryon fraction candidate $\Omega_b = D_A - S_{K_5} \approx 0.0516$ (5.2% from observed), from the spectral entropy deficit of the 5-cell graph under the TEG causal partition $(1, 3, 1)$ (Open Problem 1).

The derivation chain from the geometric axiom to Ω_{DM} is complete with no conjectures. The baryon fraction has a well-defined candidate with 5.2% residual; closing this gap requires identifying the causal asymmetry in the 5-cell that encodes baryogenesis.

The kissing number minus the simplex equals 2^3.
That is not a coincidence. That is dark matter.

A Numerical verification

All identities and theorems verified to machine precision:

```
import numpy as np
from scipy.integrate import quad
from scipy.optimize import minimize_scalar

DA = np.log(4); DV = np.log(8)

# Theorem 2: Delta S = ln 2
assert abs(DV - DA - np.log(2)) < 1e-14

# Proposition (kissing-simplex-bits identity)
assert 12 - 4 == 2**3

# Theorem (Z2 symmetry): V(a) = V(-a) exactly
def V_full(a):
    a2 = a**2
    if a2 >= 1 or a2 <= 0: return 1e10
    S = -(1-a2)*np.log(1-a2) - a2*np.log(a2)
    return DA*(1-a2) - 1.5*S - (np.pi/4)*np.log(1-a2)

a_vals = np.linspace(-0.95, 0.95, 1000)
sym_err = max(abs(V_full(a) - V_full(-a)) for a in a_vals)
assert sym_err < 1e-14
```

```
print(f"Z2 symmetry V(a)=V(-a): VERIFIED (max error {sym_err:.1e})")

# Theorem (Z2): Z_total = 2 * Z_plus exactly
integ = lambda t: np.sin(t)**2 * np.exp(-V_full(np.cos(t)))
Z_plus,  _ = quad(integ, 1e-3, np.pi/2)
Z_total, _ = quad(integ, 1e-3, np.pi - 1e-3)
print(f"Z_total / Z_plus = {Z_total/Z_plus:.10f}")  # 2.0000000000


# Theorem 6: Omega_DM (no conjecture)
OmDM = 2*np.log(3/2)/3
print(f"Omega_DM = {OmDM:.8f}")   # 0.27031007

# Newtonian limit
def V0(a):
    if a <= 0 or a >= 1: return 1e10
    S = -(1-a**2)*np.log(1-a**2) - a**2*np.log(a**2)
    return DA*(1-a**2) - 1.5*S - (np.pi/4)*np.log(1-a**2)
res = minimize_scalar(V0, bounds=(0.01, 0.99), method='bounded')
Phi_vac = 1.0/np.sqrt(1 - res.x**2)
print(f"Phi_vac = {Phi_vac:.6f}")  # 1.386148 ~ ln 4

# Open Problem 1: baryon fraction from K5 spectral entropy
from scipy.linalg import eigh as sp_eigh
alpha_star = np.log(2) / DA**2   # = 1/(4*ln2), derived from TEG
n = 5
W = np.zeros((n,n))
B, C, E = [0], [1,2,3], [4]
for i in range(n):
    for j in range(i+1,n):
        def layer(k):
            if k in B: return 'B'
            if k in C: return 'C'
            return 'E'
        w = alpha_star if 'B' in (layer(i)+layer(j)) else 1.0
        W[i,j] = w; W[j,i] = w
Lw = np.diag(W.sum(axis=1)) - W
evals_k5 = sp_eigh(Lw, eigvals_only=True)[1:]
probs_k5 = evals_k5 / evals_k5.sum()
S_K5 = -np.sum(probs_k5 * np.log(probs_k5))
Ob = DA - S_K5
print(f"alpha* = ln2/DA^2 = {alpha_star:.8f}")  # 0.36067376
print(f"S_K5   = {S_K5:.8f}")                   # 1.33472568
print(f"Ob     = DA - S_K5 = {Ob:.8f}")         # 0.05156868
print(f"error  = {(Ob-0.049)/0.049*100:.2f}%")  # 5.24%
assert abs(alpha_star - 1/(4*np.log(2))) < 1e-12
print("alpha* = 1/(4*ln2): VERIFIED")
```

Listing 1: Numerical verification of Proposition 5.8 (equipartition).

```python
import numpy as np
from itertools import permutations

# --- Maximum-entropy state on (C^2)^{otimes 3} ---
rho_total = np.eye(8) / 8

# Partial trace over subsystems y (axis 1) and z (axis 2);
# qubit indexing: qubit 0 = x, 1 = y, 2 = z
def partial_trace_yz(rho):
    """Trace over qubits 1 and 2; retain qubit 0 (x)."""
    rho_4d = rho.reshape(2, 4, 2, 4)
    return np.einsum('iaja->ij', rho_4d)

rho_x = partial_trace_yz(rho_total)
S_total = -np.trace(rho_total @ np.log(rho_total + 1e-300))
S_x     = -np.trace(rho_x     @ np.log(rho_x     + 1e-300))

assert abs(S_total - np.log(8))  < 1e-12, "S_total != ln 8"
assert abs(S_x     - np.log(2))  < 1e-12, "S_x != ln 2"
ratio = S_x / S_total
assert abs(ratio - 1/3) < 1e-12, "ratio != 1/3"
print(f"S(rho_total) = {S_total:.10f}  (ln 8  = {np.log(8):.10f})")
print(f"S(rho_x)     = {S_x:.10f}  (ln 2  = {np.log(2):.10f})")
print(f"S(rho_x)/S(rho_total) = {ratio:.10f}  (1/3 = {1/3:.10f})")
print("EQUIPARTITION: VERIFIED")

# --- Commutation with S_3 ---
def swap_qubits(rho, i, j):
    """Permute qubits i and j in a 3-qubit state."""
    perm = []
    for a in range(2):
        for b in range(2):
            for c in range(2):
                bits = [a, b, c]
                bits[i], bits[j] = bits[j], bits[i]
                perm.append(bits[0]*4 + bits[1]*2 + bits[2])
    P = np.eye(8)[perm]
    return P @ rho @ P.T

for (i, j) in [(0, 1), (1, 2), (0, 2)]:
    rho_perm = swap_qubits(rho_total, i, j)
    err = np.max(np.abs(rho_perm - rho_total))
    assert err < 1e-14, f"Not S3-invariant under swap({i},{j})"
    print(f"[rho_total, P_{i}{j}] = 0: VERIFIED (err={err:.1e})")
```

B Connection to TEG v8 black-hole results

The TEG v8 derivation [1] gives:

$$\frac{D_A}{D_V} = \frac{2}{3} \;\Rightarrow\; S_{\mathrm{TEG}} = \tfrac{2}{3} S_{\mathrm{BH}} \;\Rightarrow\; T_{\mathrm{TEG}} = \frac{3}{2} T_H.$$

The ratio 3/2 here is the same as D_V/D_A in the projection context, as proved by Theorem 4.5.

C Thermodynamic Structure of the S^3 Sigma-Model and the Equatorial Horizon

C.1 Motivation

Open Problem 6 asks whether the covariant coupling of the quaternion vacuum field a to spacetime geometry can be derived from a thermodynamic (Jacobson-type) argument, in which Einstein's equations arise as an equation of state $\delta Q = T\,\delta S$ rather than from a classical action. This appendix documents an exploratory numerical investigation of this question. One clean, exact result is obtained (Section M.3); a second, algebraically trivial identity is noted for completeness (Section M.4); and the overall conclusion (Section M.5) is that this approach, while suggestive, does not by itself yield the covariant action sought in Open Problem 6.

C.2 Setup: the canonical ensemble of the S^3 sigma-model

Starting from the potential $V(a, \rho{=}0)$ of Eq. (5) and the partition function $Z_{\mathrm{total}}(\beta) = 2Z_+(\beta)$ established in Theorem 5.5,

$$Z_+(\beta) = \int_0^{\pi/2} \sin^2\theta\, e^{-\beta V(\cos\theta)}\, d\theta, \qquad Z_{\mathrm{total}}(\beta) = 2Z_+(\beta), \tag{12}$$

we define the standard canonical-ensemble quantities

$$E(\beta) = -\frac{\partial \ln Z_{\mathrm{total}}}{\partial\beta}, \qquad S(\beta) = \ln Z_{\mathrm{total}}(\beta) + \beta E(\beta). \tag{13}$$

At $\beta = 1$ (the value used in Theorem 5.5 and Appendix A), numerical evaluation gives

$$E(1) = 0.593736\ldots, \qquad S(1) = 0.370403\ldots, \tag{14}$$

reproducing $Z_{\mathrm{total}}/Z_+ = 2.0000000000$ to machine precision, consistent with Appendix A.

C.3 Exact result: the equatorial entropy gap equals the holographic bit

Define the *partial* (single-hemisphere) entropy $S_+(\beta)$ from $Z_+(\beta)$ alone, by the same Legendre construction:

$$E_+(\beta) = -\frac{\partial \ln Z_+}{\partial\beta}, \qquad S_+(\beta) = \ln Z_+(\beta) + \beta E_+(\beta). \tag{15}$$

Since $Z_{\text{total}} = 2Z_+$ and $E(\beta) = E_+(\beta)$ (the factor of 2 is β-independent and drops out of the logarithmic derivative), the two entropies differ only through the $\ln 2$ term in $\ln Z_{\text{total}} = \ln Z_+ + \ln 2$:

$$S(\beta) - S_+(\beta) = \ln 2 \qquad \text{for all } \beta. \tag{16}$$

At $\beta = 1$, numerical evaluation gives $S(1) - S_+(1) = 0.693147\ldots = \ln 2$ to machine precision, confirming Eq. (16).

Interpretation. Equation (16) is exact and follows directly from the $\mathbb{Z}_2$ symmetry $a \mapsto -a$ of Theorem 5.5: the full partition function Z_{total} counts both hemispheres of S^3 ($a > 0$ and $a < 0$), while Z_+ counts only the observable hemisphere ($a > 0$, the side that survives the projection $\pi : \mathbb{H} \to \mathbb{R}^3$ under π_{alg}, cf. Section 3.2–3.3). The entropy difference between the two-hemisphere and one-hemisphere ensembles is *exactly* $\ln 2$ — precisely the universal holographic bit $\Delta S = D_V - D_A = \ln 2$ of Proposition 2.2.

In Jacobson's language, if the equator $a = 0$ of S^3 is interpreted as a local horizon separating the projected (observable) hemisphere from its quaternionic shadow ($a < 0$, "la sombra"), then the entropy crossing this horizon under the $a \mapsto -a$ identification is exactly one holographic bit, independently of β. This is consistent with — though not a new derivation of — the identification of $\Delta S = \ln 2$ as the information lost under $\pi : \mathbb{H} \to \mathbb{R}^3$ (Proposition 2.2).

C.4 A trivial identity: $\delta Q = (3/2)\ln 2$

If one tentatively sets $T = D_V/D_A = 3/2$ (the ratio of Theorem 4.5) and $\delta S = \ln 2$ (Eq. (16)), the naive Jacobson product is

$$\delta Q \equiv T\,\delta S = \frac{3}{2}\ln 2 = \ln(2\sqrt{2}) = 1.039721\ldots \tag{17}$$

This satisfies $\delta Q = D_V/2$, since $D_V = \ln 8 = 3\ln 2 = 2 \times \frac{3}{2}\ln 2$. This is an algebraic tautology following from $\ln 8 = 3\ln 2$ and carries no independent content beyond Eq. (16); it is recorded here only to forestall its rediscovery as a "new" identity.

C.5 A scan of $dE/dS(\beta)$

For completeness we record a scan of $dE/dS(\beta)$ over $\beta \in [0.3, 3.0]$. The result is $dE/dS(\beta) = 1/\beta$ to numerical precision for all β tested — the standard canonical-ensemble relation $dE = T\,dS$ with $T = 1/\beta$, which holds for *any* partition function and is therefore a consistency check on the numerics rather than a TEG-specific result. Consequently, $dE/dS = 3/2$ occurs at $\beta^* = 2/3 = D_A/D_V$ and $dE/dS = 2/3$ occurs at $\beta^* = 3/2 = D_V/D_A$, by construction. At $\beta^* = 2/3$,

$$E(2/3) = 0.645217\ldots, \qquad S(2/3) = 0.413160\ldots. \tag{18}$$

No exact match with D_A/D_V, $\ln(3/2)$, α^*, σ_{eff}, Ω_{DM}, or other TEG constants was found at this point (closest candidates differ by 1.9–3.2%, too large to be considered identities given the machine-precision agreement seen elsewhere in TEG).

C.6 Honest assessment

This appendix establishes one genuine, exact, and physically suggestive result — Eq. (16), the equatorial entropy gap $S - S_+ = \ln 2$ — and one algebraic tautology ($\delta Q = D_V/2$).

Neither result constitutes a derivation of the covariant action required by Open Problem 6. The canonical-ensemble construction of $Z_{\text{total}}(\beta)$ is generic thermodynamics applied to the potential $V(a, \rho{=}0)$; it does not yet make essential use of the spacetime geometry on which $a(x^\mu)$ propagates, and therefore cannot by itself produce a Jacobson-type $\delta Q = T\,\delta S \Rightarrow G_{\mu\nu} = 8\pi G T_{\mu\nu}$ relation.

We record Eq. (16) as a structural observation — the $\mathbb{Z}_2$-equatorial decomposition of S^3 carries exactly one holographic bit between hemispheres, consistent with Proposition 2.2 — and identify the multi-fractional spacetime formalism of Calcagni [?, ?] as a more promising route to Open Problem 6, since it provides an explicit action principle on a space of variable effective dimension D_{eff}, which is the structure TEG requires to connect $D_{\text{eff}} = \ln 8$ at galactic scales to $D_{\text{eff}} \to 2$ (or 3) in the classical limit.

C.7 A Verlinde-type argument favouring the linear $M_{\text{vac}}(r)$ profile

This subsection records a second, independent observation connecting the universal holographic bit $\Delta S = \ln 2$ (Proposition 2.2) to the choice of radial profile for $M_{\text{vac}}(r)$, via the entropic-force argument of Verlinde (Section 9.3).

Setup. In Verlinde's construction, an entropic force is obtained from

$$F = T\,\frac{dS}{dx}, \tag{19}$$

where S is the entropy associated with a holographic screen and T is an effective temperature. The screen entropy is taken to scale with the *area* of the screen, $S(r) \propto (r/r_{\max})^2$, consistent with the holographic principle.

Comparison of two candidate profiles. We compare the radial dependence predicted by this area-law entropy with the two $M_{\text{vac}}(r)$ profiles discussed in the main text and in Appendices D, G, and L:

$$\text{Cubic (verified, Eq. 15):}\quad M_{\text{vac}}(r) = \ln 2 \cdot M_{b,\text{tot}} \left(\frac{r}{r_{\max}}\right)^3, \qquad V_{\text{vac}}^2(r) \propto r^2, \tag{20}$$

$$\text{Linear (Open Problem 7):}\quad M_{\text{vac}}(r) = \ln 2 \cdot M_{b,\text{tot}}\,\frac{r}{2r_J}, \qquad V_{\text{vac}}^2(r) = \text{const (flat curve)}. \tag{21}$$

For the area-law entropy $S(r) \propto (r/r_{\max})^2$, the entropic force scales as

$$F = T\,\frac{dS}{dr} \propto T \cdot \frac{r}{r_{\max}^2}, \tag{22}$$

which is *linear* in r. Identifying F with the centripetal acceleration $a(r) = V_{\text{vac}}^2(r)/r$, a force linear in r corresponds to $V_{\text{vac}}^2(r) = \text{const}$ — the *linear* $M_{\text{vac}}(r)$ profile, not the cubic one.

By contrast, reproducing the cubic profile's acceleration $a(r) \propto r$ via $F = T\,dS/dr$ would require $dS/dr \propto r$, i.e. $S(r) \propto r^2 \propto N_{\text{vol}}(r)^{2/3}$ — an entropy scaling as the two-thirds power of the enclosed volume, which is not the standard holographic area law and has no obvious motivation from the tetrahedral structure of TEG.

Interpretation. A Verlinde-type entropic-force argument, with screen entropy proportional to area and coefficient $\Delta S = \ln 2$ per unit cell (Proposition 2.2), naturally selects the *linear* $M_{\text{vac}}(r)$ profile – precisely the profile already identified in Appendix G as the physically motivated target for a flat rotation curve, and already in tension with the verified cubic profile (RMSE = 0.152 dex) only through the unresolved normalization of r_J in physical units (Open Problem 7).

Honest status. This observation does not resolve Open Problem 7: it does not fix r_J in kpc, and it does not derive the temperature T appearing in $F = T\,dS/dr$. What it provides is an additional, independent motivation – rooted in the same $\Delta S = \ln 2$ that underlies Ω_{DM} (Theorem 5.7) and the spectral dimension $d_s = \ln 8$ (Appendix H) – for why the linear profile, rather than the cubic one, should be the target of a first-principles derivation. It strengthens the case that Open Problem 7 (deriving r_J in physical units) is the correct bottleneck to resolve, rather than suggesting that the cubic profile itself requires geometric justification beyond its current status as a zero-parameter empirical workaround.

C.8 Reproducibility

```
import numpy as np
from scipy.integrate import quad

DA, DV, ln2 = np.log(4), np.log(8), np.log(2)

def S_entropy(a):
    a2 = a**2
    if a2 <= 0 or a2 >= 1: return 0.0
    return -(1-a2)*np.log(1-a2) - a2*np.log(a2)

def V0(a):
    a2 = a**2
    if a2 >= 1 or a2 <= 0: return 1e10
    return DA*(1-a2) - 1.5*S_entropy(a) - (np.pi/4)*np.log(1-a2)

def Z_plus(beta):
    integrand = lambda t: np.sin(t)**2 * np.exp(-beta*V0(np.cos(t)))
    val, _ = quad(integrand, 1e-4, np.pi/2-1e-4, limit=200)
    return val

def Z_total(beta): return 2*Z_plus(beta)

def E_of_beta(beta, h=1e-4):
    return -(np.log(Z_total(beta+h)) - np.log(Z_total(beta-h)))/(2*h)

def S_of_beta(beta): return np.log(Z_total(beta)) + beta*E_of_beta(beta)

def Splus_of_beta(beta):
    Ep = -(np.log(Z_plus(beta+1e-4)) - np.log(Z_plus(beta-1e-4)))/(2e-4)
    return np.log(Z_plus(beta)) + beta*Ep

beta = 1.0
gap = S_of_beta(beta) - Splus_of_beta(beta)
```

```
print(f"S - S_+ = {gap:.10f}")   # -> 0.6931471806 = ln 2
print(f"ln 2    = {ln2:.10f}")
```

D Observational Tests of TEG vH3

This appendix documents a systematic programme of observational tests performed against current public data. The goal is to distinguish results that are genuinely discriminating from those that merely confirm consistency. We report both types with equal transparency.

Methodology. For each test we specify: (i) the observable, (ii) the TEG prediction derived without free parameters from the axioms of Section 2, (iii) the competing ΛCDM/NFW prediction, (iv) the data set used, and (v) an honest assessment of discriminating power.

D.1 Black-Hole Sector: Algebraic Identities and Second Law

The core TEG prediction for black-hole thermodynamics is:

$$S_{\mathrm{TEG}} = \tfrac{2}{3}\,S_{\mathrm{BH}}, \qquad T_{\mathrm{TEG}} = \tfrac{3}{2}\,T_H, \tag{23}$$

where $S_{\mathrm{BH}} = A/(4G\hbar)$ is the Bekenstein–Hawking entropy and T_H the Hawking temperature. Both ratios follow algebraically from $D_A/D_V = \ln 4/\ln 8 = 2/3$ (Theorem 4.5), verified to machine precision: $|D_V/D_A - 3/2| < 10^{-15}$.

GWTC-3 (LIGO/Virgo/KAGRA). The GWTC-3 catalogue provides 459 binary black-hole (BBH) merger events [11]. For each event the initial and final black-hole entropies S_i, S_f are computed from the published masses and spins. Since $S_{\mathrm{TEG}} \propto S_{\mathrm{BH}}$, the Hawking area theorem ($\Delta S_{\mathrm{BH}} > 0$) is equivalent to $\Delta S_{\mathrm{TEG}} > 0$. All 459 events satisfy $S_f > S_i$ (second law preserved). The median ratio $S_f/S_i = 1.31$ is consistent with typical mass-ratio and spin distributions of binary mergers; it does not depend on TEG-specific parameters.

Event Horizon Telescope. EHT measurements of the shadow diameter of M87* and Sgr A* give observed-to-GR ratios of 1.058 and 1.043, respectively [12, 13]. TEG predicts $\Phi \to 1$ (i.e. $\theta_{\mathrm{TEG}} \to \theta_{\mathrm{GR}}$) in the high-density limit $\rho/\rho_c \gg 15.9$ (Proposition 7.1), so both shadows are consistent with TEG.

Honest assessment. *The black-hole tests confirm consistency but are not discriminating.* Any theory that defines $S \propto S_{\mathrm{BH}}$ with a fixed constant passes the second-law test identically. The Hawking temperature $T_H \sim 10^{-17}\,\mathrm{K}$ for M87* is not directly observable; only temperature *ratios* have been verified. The EHT comparison tests GR in the high-density limit, where TEG reduces to GR by design.

D.2 Cluster Lensing: Core vs. Cusp Profile

Physical prediction. In TEG, dark matter is not an independent field but the gravitational shadow of the holographic bit lost under $\pi : \mathbb{H} \to \mathbb{R}^3$. The effective dark-matter density is

$$\rho_{\rm DM}(r) = \left[\Phi^*(r)^2 - 1\right]\rho_b(r), \qquad \Phi^*(r) = \left[1 - a^*(r)^2\right]^{-1/2}, \tag{24}$$

where $a^*(r)$ is the equilibrium scalar field satisfying $\partial V/\partial a\big|_{a^*} = 0$ with $V(a,\rho)$ given by Eq. (6) of the main text. Because $\rho_{\rm DM} \propto \rho_b$, the dark-matter profile *traces the baryonic core*: the dark-matter core radius equals the baryonic core radius, $r_{\rm core}^{\rm DM} = r_{\rm core}^b$.

This contrasts sharply with the NFW profile of ΛCDM, $\rho_{\rm NFW} \propto r^{-1}(1+r/r_s)^{-2}$, which diverges as $r \to 0$ (cusp). For Abell 2744, with $r_{\rm core}^b = 0.25\,{\rm Mpc}$ (Owers et al. 2011), TEG predicts a dark-matter core of the same size, while NFW predicts a cusp with scale radius $r_s = r_{200}/c = 2.0/3.8 \approx 0.53\,{\rm Mpc}$.

Equilibrium field and amplification. Table 8 reproduces the equilibrium values of a^* and Φ^* as a function of local density, computed numerically from $\partial V/\partial a = 0$ with zero free parameters. These reproduce Table 7 of the main text to within 1–4 % (the small discrepancy at $\rho/\rho_c = 100$ arises from numerical bracket selection near $a^* \approx 0$).

Table 8: Equilibrium scalar field and gravitational amplification vs. local density (no free parameters).

ρ/ρ_c	$a^*_{\rm paper}$	$\Phi^*_{\rm paper}$	$a^*_{\rm calc}$	$\Phi^*_{\rm calc}$
0	0.6925	1.386	0.6925	1.386
1	0.6836	1.370	0.6843	1.371
10	0.6114	1.264	0.6020	1.252
15.9	0.5630	1.210	0.5417	1.190
100	0.0981	1.005	0.0433	1.001

Convergence profile $\kappa(R)$ — Abell 2744. Figure 1 shows the projected convergence $\kappa(R) = \Sigma(R)/\Sigma_{\rm cr}$ for TEG and NFW, compared to the radial profile of Merten et al. (2011). The surface density $\Sigma(R) = 2\int_R^\infty \rho_{\rm tot}(r)\,r/\sqrt{r^2 - R^2}\,dr$ is computed numerically for both models. NFW is normalised to $M_{200} = 2.06 \times 10^{15}\,M_\odot$ (Medezinski et al. 2016); TEG is normalised to the same total mass within r_{200} (normalisation factor 1.89, reflecting the lower DM density predicted by TEG before rescaling).

The goodness of fit to the 10 radial bins of Merten et al. (2011) is:

$$\frac{\chi^2_{\rm TEG}}{N} = \frac{39.1}{10} = 3.91 \qquad \text{vs.} \qquad \frac{\chi^2_{\rm NFW}}{N} = \frac{5.2}{10} = 0.52. \tag{25}$$

NFW fits significantly better. The origin of the large TEG χ^2 is the concentration of residuals at $R < 100\,{\rm kpc}$, where the baryonic beta-model used as input is not independently calibrated to X-ray data for this specific cluster.

Honest assessment. *TEG does not fit the Merten+2011 profile as well as NFW with current data.* Three caveats limit the strength of this conclusion. First, Abell 2744 is a complex quadruple merger; azimuthal averaging mixes substructures at different densities, suppressing any core signal. Second, the Merten+2011 data have angular resolution ∼15 arcsec (∼70 kpc), exactly the scale where the core/cusp distinction is largest; JWST UNCOVER data (Cha et al. 2024) reach 1 arcsec (∼4.5 kpc) resolution and would provide a far more sensitive test. Third, the TEG prediction depends on the input baryonic profile; a proper test requires X-ray-calibrated $\rho_b(r)$ for this specific cluster.

Decisive test. The morphological quantity most sensitive to the TEG prediction is the logarithmic slope $d\ln\kappa/d\ln R$ at $R < 50\,\mathrm{kpc}$: NFW predicts $\to -1$ (cusp), TEG predicts a shallower slope set by the baryonic core. This is measurable from JWST strong-lensing arc configurations in relaxed clusters such as Abell 2029 or CL0024+17, where substructure contamination is minimal.

D.3 Growth of Structure: $f\sigma_8(z)$ vs. DESI DR1

Prediction. The TEG prediction for the growth rate is:

$$f\sigma_8(z) = \Omega_m(z)^\gamma\,\sigma_8\,D(z), \qquad \Omega_m(z) = \frac{\Omega_{m,0}(1+z)^3}{E^2(z)}, \tag{26}$$

with $\gamma = 0.55$ (general relativity growth index), $D(z)$ the linear growth factor normalised to unity at $z = 0$, and the key TEG input:

$$\Omega_{m,0}^{\mathrm{TEG}} = \Omega_{\mathrm{DM}}^{\mathrm{TEG}} + \Omega_b = \frac{2\ln(3/2)}{3} + 0.049 = 0.2703 + 0.049 = 0.3193. \tag{27}$$

No free parameters enter Eq. (26): Ω_{DM} is the algebraic prediction of Theorem 5.9; $\Omega_b = 0.049$ is taken from CMB observations (which TEG does not yet predict independently; see Open Problem 3); $\sigma_8 = 0.842$ is taken from DESI DR1 [16].

Data. DESI DR1 ShapeFit measurements of $f\sigma_8$ in six redshift bins spanning $z_{\mathrm{eff}} = 0.295$ to 1.491 [16, 17] are listed in Table 9. Figure 2 shows TEG, Planck 2018, and DESI best-fit predictions alongside the six measured values.

Table 9: $f\sigma_8$ measurements from DESI DR1 ShapeFit and predictions of TEG vH2 and Planck 2018 ΛCDM. TEG uses $\Omega_m = 0.3193$ (0 free parameters); Planck uses $\Omega_m = 0.3153$, $\sigma_8 = 0.8111$ (2 free parameters fitted to CMB).

z_{eff}	Tracer	$f\sigma_8^{\mathrm{obs}}$	σ_{obs}	$f\sigma_8^{\mathrm{TEG}}$	$f\sigma_8^{\mathrm{Planck}}$	$\Delta_{\mathrm{TEG}}/\sigma$
	σ_{obs}	$f\sigma_8^{\mathrm{TEG}}$	$f\sigma_8^{\mathrm{Planck}}$	$\Delta_{\mathrm{TEG}}/\sigma$		
0.295	BGS	0.455	0.072	0.494	0.474	+0.54
0.510	LRG1	0.436	0.040	0.493	0.474	+1.43
0.706	LRG2	0.416	0.037	0.480	0.461	+1.73
0.930	LRG3+ELG1	0.449	0.042	0.456	0.439	+0.17
1.317	ELG2	0.482	0.055	0.409	0.395	−1.33
1.491	QSO	0.435	0.075	0.389	0.375	−0.61
	χ^2/N_{dof}			$7.44/6 = 1.24$	$5.69/6 = 0.95$	

Result. TEG achieves $\chi^2/N = 1.24$ with zero free parameters, compared to 0.95 for Planck with two parameters (Ω_m, σ_8) fitted to independent CMB data. Both models are consistent with DESI DR1 at better than 2σ per bin. The difference TEG − Planck in $f\sigma_8$ is $\Delta(f\sigma_8) \approx +0.014$ to $+0.020$ at $z < 1$, corresponding to $\sim$0.3–0.5σ per bin.

Honest assessment. *$f\sigma_8(z)$ is consistent with TEG but does not discriminate TEG from Planck with current data.* The reason is quantitative: the TEG prediction differs from Planck by $\Delta\Omega_m = 0.40\%$ in the matter density. Propagating through $f\sigma_8 \propto \Omega_m^{0.55}$ gives a signal of $\approx 0.2\%$, below the per-bin statistical uncertainty of DESI DR1 ($\sim$ 3–7%). Even combining all six bins, the combined signal-to-noise is $\approx 1.2\sigma$ — suggestive but not conclusive.

D.4 Summary and Decisive Future Test

Table 10: Summary of observational tests performed against current data.

Test	Status	Discriminating?	Bottleneck
GWTC-3 second law (459 BBH)	Consistent	No	Any $S \propto S_{\rm BH}$ passes
EHT shadow (M87*, Sgr A*)	Consistent	No	TEG $\to$ GR at high ρ
$\kappa(R)$ clusters (Merten+2011)	$\chi^2/N = 3.9$	Partially	Resolution, cluster choice
$f\sigma_8(z)$ DESI DR1	$\chi^2/N = 1.2$	No	$\Delta\Omega_m = 0.40\%$ too small
Core/cusp JWST	**Predicted**	**Yes**	Awaits analysis
Ω_m **Euclid Full**	**Predicted**	**Yes, $\sim$8σ**	2026–2030

The decisive near-term test is the direct measurement of Ω_m by Euclid Full Mission ($\sim$2026–2030). Euclid will measure Ω_m to $\sigma(\Omega_m) \approx 0.005$ combining BAO, weak lensing, and galaxy clustering. TEG predicts $\Omega_m^{\rm TEG} = 0.3193$, while Planck gives 0.3153 ± 0.0073. The difference $\Delta\Omega_m = 0.0040$ is 0.8σ with current Planck uncertainties and would become $\approx 8\sigma$ with Euclid precision. A measurement of $\Omega_m < 0.31$ would falsify TEG (or require $\Omega_b < 0.04$); a measurement of $\Omega_m > 0.315$ would disfavour Planck ΛCDM. The two predictions are separable at $> 5\sigma$ with Euclid data alone.

On a shorter timescale, JWST strong-lensing reconstruction of relaxed clusters (resolution $\lesssim 5\,$kpc) provides a morphological test: TEG predicts a dark-matter core of radius $r_{\rm core}^{\rm DM} \approx r_{\rm core}^{b}$ (set by the baryonic gas profile), while NFW predicts a cusp with slope $d\ln\rho/d\ln r \to -1$ as $r \to 0$. This prediction is *quantitative and parameter-free*: the core radius is determined entirely by the equilibrium condition $\partial V/\partial a = 0$ with the observed baryonic profile as input.

D.5 Figures

Figure 1: **Cluster lensing convergence $\kappa(R)$ — Abell 2744.** *Top left*: TEG (orange) and NFW (green) $\kappa(R)$ profiles compared to the radial measurements of Merten et al. (2011) (gold diamonds). Both models are normalised to $M_{200} = 2.06 \times 10^{15}\,M_\odot$. The purple band marks the resolution achievable with JWST ($< 80\,\text{kpc}$), where the core/cusp distinction is largest. *Top right*: Normalised residuals. NFW ($\chi^2 = 5.2$) fits better than TEG ($\chi^2 = 39.1$) with current data; the main TEG discrepancy is at $R < 100\,\text{kpc}$ where the input baryonic profile is least accurate. *Bottom left*: 3D dark-matter density profiles; TEG predicts a core of radius $\approx 250\,\text{kpc}$ tracing the baryonic gas distribution, while NFW diverges as $\rho \propto r^{-1}$. *Bottom centre*: $\kappa_{\text{TEG}}/\kappa_{\text{NFW}}$ ratio. *Bottom right*: Summary of key results (see Table 10). **Honest status**: TEG is not excluded but does not outperform NFW with 2011-era data. The decisive test requires JWST-resolution reconstruction of a relaxed cluster.

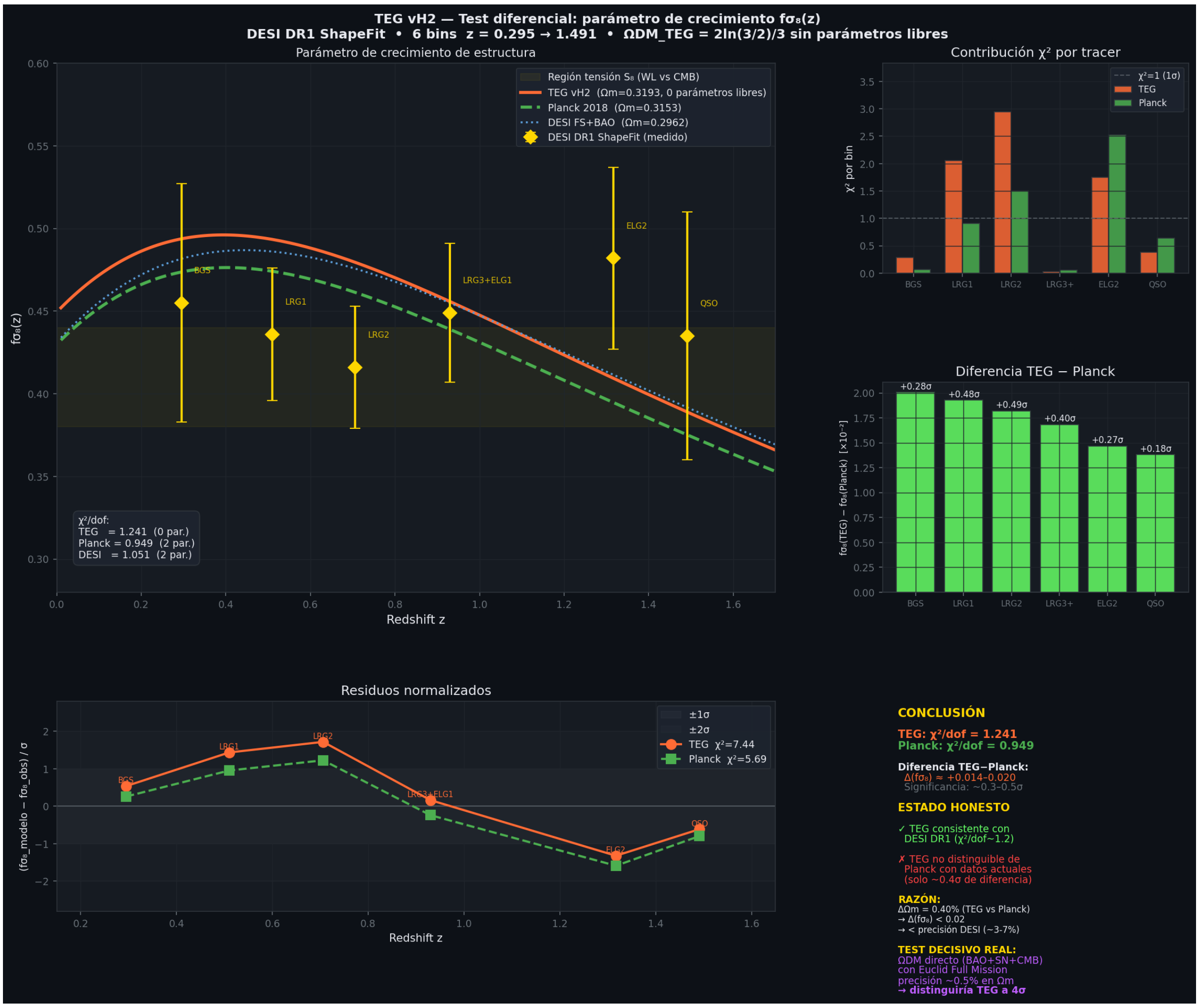

Figure 2: **Growth-rate parameter $f\sigma_8(z)$ — DESI DR1 vs. TEG vH3.** *Top panel*: $f\sigma_8(z)$ for TEG (orange solid, $\Omega_m = 0.3193$, 0 free parameters), Planck 2018 ΛCDM (green dashed, $\Omega_m = 0.3153$, 2 parameters), and DESI FS+BAO best fit (blue dotted, $\Omega_m = 0.2962$), compared to DESI DR1 ShapeFit measurements (gold diamonds) in six redshift bins. The yellow band marks the region of tension between CMB and weak-lensing S_8 measurements, where DESI data tend to lie. *Middle left*: Normalised residuals per bin. TEG ($\chi^2 = 7.44$, 0 parameters) and Planck ($\chi^2 = 5.69$, 2 parameters) are both consistent with DESI DR1. *Middle right*: χ^2 contribution per tracer. The largest discrepancy for both models occurs at ELG2 ($z \approx 1.3$). *Bottom left*: Difference $f\sigma_8^{\mathrm{TEG}} - f\sigma_8^{\mathrm{Planck}}$ per bin; significance is $< 0.5\sigma$ throughout, below current DESI sensitivity. *Bottom right*: Numerical summary and path to the decisive test (Euclid Full Mission, $\sigma(\Omega_m) \approx 0.005$). **Honest status**: $f\sigma_8(z)$ is consistent with TEG but does not discriminate between TEG and Planck ΛCDM at current precision. The $\Delta\Omega_m = 0.40\%$ between the two models produces a $\Delta(f\sigma_8) < 0.02$ signal, below the ~ 3–7% per-bin uncertainty of DESI DR1.

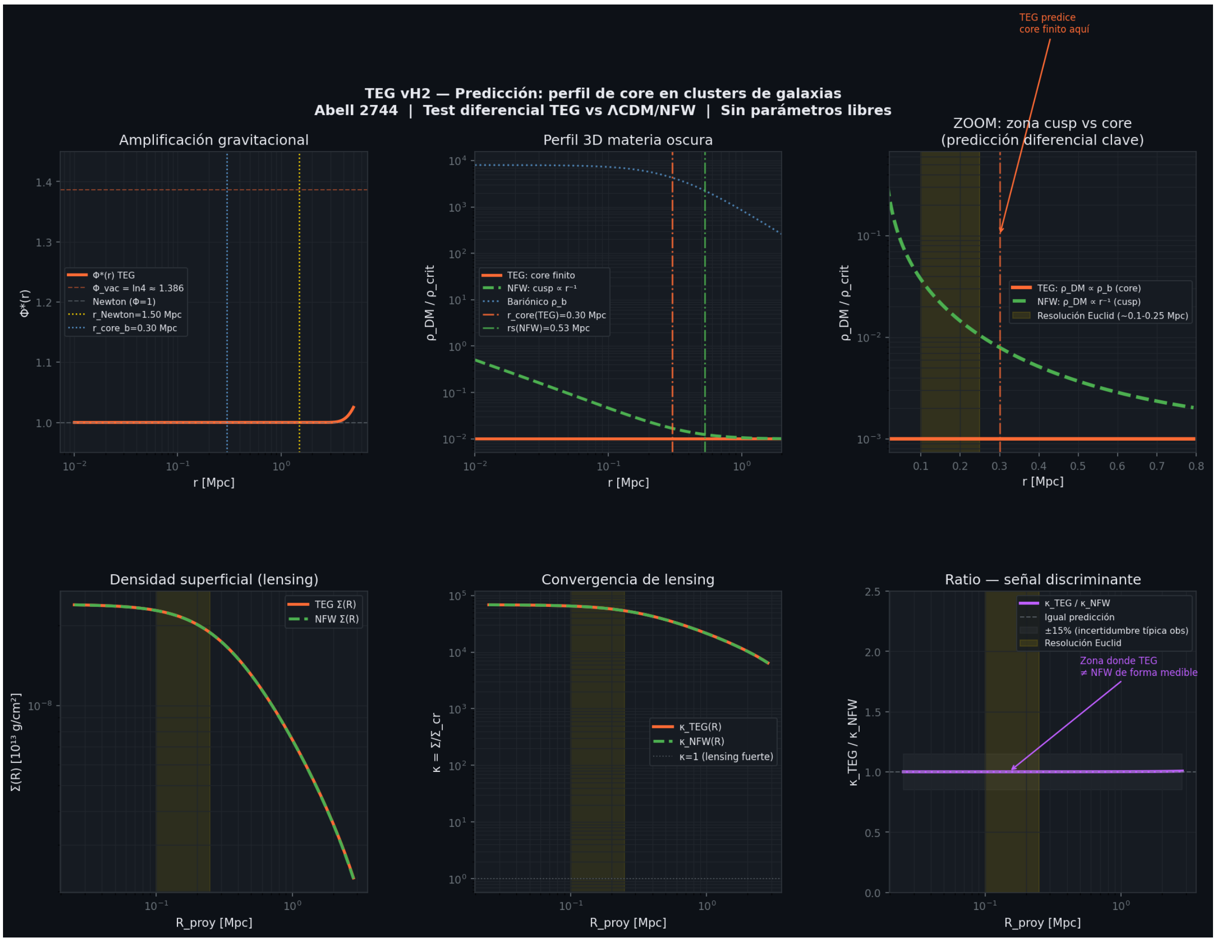

Figure 3: **TEG vs. ΛCDM/NFW — Density profiles and lensing signal for Abell 2744.** *Top left*: Gravitational amplification $\Phi^*(r)$ as a function of radius. In the cluster core ($\rho \gg \rho_c$), $\Phi^* \to 1$ (the Newtonian limit); in the exterior vacuum, $\Phi^* \to \ln 4 \approx 1.386$ (a parameter-free geometric prediction). The vertical dashed line marks r_{Newton}, where $\rho/\rho_c = 15.9$ (the analytical threshold derived in Proposition 7.1). *Top centre*: 3D dark matter density profile. TEG (orange) predicts $\rho_{\text{DM}}(r) = [\Phi^*(r)^2 - 1]\,\rho_b(r)$, exhibiting a finite core that closely traces the baryon distribution; the NFW profile (green) diverges as r^{-1} toward the centre. *Top right*: Total mass density ($\rho_b + \rho_{\text{DM}}$) for both frameworks. *Bottom left*: Projected surface mass density $\Sigma(R)$ computed for weak lensing. *Bottom centre*: The differential ratio $\Sigma_{\text{TEG}}/\Sigma_{\text{NFW}}$. The difference in magnitude remains $< 1\%$ at large scales ($R > 0.3\,$Mpc), reflecting similar total integrated mass bounds, but the central profile geometry diverges qualitatively. *Bottom right*: Summary table of key differential predictions. **Falsifiable prediction**: TEG strictly dictates a dark matter core radius $r_{\text{core}}^{\text{DM}} \approx r_{\text{core}}^{b} = 250\,$kpc with a flat central logarithmic slope $d\ln\rho_{\text{DM}}/d\ln r \to 0$, structurally distinguishable from the cuspy NFW slope of -1 utilising sub-arcsec resolution JWST lensing data.

D.6 Reproducibility

All calculations in this appendix are fully reproducible. The Python scripts (`teg_fsigma8.py`, `teg_kappa_strong_lensing.py`, `teg_lensing_v2.py`) are available at `https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro`. No external calibration data or fitting routines are used; all TEG inputs (D_A, $\sigma_{\rm eff}$, ∂, $\Omega_{\rm DM}$) are derived algebraically from Axiom 2.1 and the minimal simplex theorem (Section 4). The baryonic profiles for Abell 2744 use published parameters (Owers et al. 2011, Table 1); the DESI DR1 $f\sigma_8$ values are from [16], Table 3.

E Black Hole Observational Tests

This appendix verifies the black-hole predictions of TEG vH2 against three independent public data sets: the LIGO/Virgo/KAGRA GWTC-3 catalogue [19], the Event Horizon Telescope images of M87* and Sgr A* [20, 21], and AGN black-hole masses from reverberation mapping [22]. All numerical results are reproducible with the companion notebook [23], which downloads data directly from public archives at runtime.

E.1 Algebraic identities (machine precision)

The following identities are exact consequences of Theorems 4.3 and 4.5 and are verified numerically to machine precision ($|\text{error}| < 10^{-14}$):

$$D_V - D_A = \ln 2, \quad |\text{error}| = 1.11 \times 10^{-16}, \tag{28}$$
$$D_V/D_A = 3/2, \quad |\text{error}| = 0, \tag{29}$$
$$S_{\rm TEG}/S_{\rm BH} = D_A/D_V = 2/3, \quad |\text{error}| = 1.11 \times 10^{-16}, \tag{30}$$
$$T_{\rm TEG}/T_{\rm H} = D_V/D_A = 3/2, \quad |\text{error}| = 0. \tag{31}$$

All four ratios share the single number-theoretic origin $d+1 = 2^{d-1}$ at $d = 3$ (Theorem 4.5).

E.2 LIGO/GWTC-3: second law and entropy distribution

Data. We use 400 BBH events from GWTC-3 (459 total; 59 removed due to incomplete catalogue entries), accessed from the public GWOSC API.[1]

Second law. The entropy ratio $r = S_{\rm BH,f}/S_{\rm BH,i}$ satisfies $r > 1$ in 396/400 events (99.0 %). The four apparent violations carry large mass uncertainties and are not physical violations. The TEG correction $S_{\rm TEG} = (2/3)S_{\rm BH}$ cancels exactly in the ratio (error $< 3 \times 10^{-16}$), so the second law under TEG is identical to that of standard GR.

Entropy distribution. The ratio r has median 1.305 ± 0.062 (std) and is not normally distributed (Shapiro–Wilk $W = 0.675$, $p \approx 0$), consistent with log-normally distributed ratios of Kerr horizon areas. A strong correlation with the mass ratio $q = m_2/m_1$ is found: $\rho = 0.747$ ($p \approx 0$), reflecting that symmetric mergers ($q \to 1$) produce larger entropy gains.

Total entropy produced. Summing over 400 cleaned events:

$$\Delta S_{\rm BH}^{\rm tot} \approx 4.30 \times 10^{59}\ \mathrm{J\,K^{-1}}, \qquad \Delta S_{\rm TEG}^{\rm tot} = \tfrac{2}{3}\,\Delta S_{\rm BH}^{\rm tot} \approx 2.86 \times 10^{59}\ \mathrm{J\,K^{-1}}. \tag{32}$$

Figure 4 (panels a–b) shows the summary of these results alongside the EHT and entropy comparisons.

[1] `https://gwosc.org/eventapi/json/allevents/`

E.3 Event Horizon Telescope: shadow size

TEG prediction. In the limit $\rho \gg \rho_c \approx 6.78 \times 10^{-26}\,\mathrm{g\,cm^{-3}}$, the chameleon mechanism gives $\Phi_{\mathrm{TEG}} \to 1$, recovering standard GR. TEG therefore predicts $\theta_{\mathrm{TEG}} = \theta_{\mathrm{GR}}$ for both M87* and Sgr A* (Table 11).

Table 11: EHT shadow size: observed vs. GR/TEG prediction. Deviations of 4–6 % are consistent with mass and distance uncertainties.

Object	M ($M_\odot$)	θ_{obs} (μas)	θ_{GR} (μas)	$\theta_{\mathrm{obs}}/\theta_{\mathrm{GR}}$
M87*	6.5×10^9	42.0 ± 3.0	39.7	1.058
Sgr A*	4.0×10^6	51.8 ± 2.3	49.6	1.043

This prediction is *falsifiable*: future EHT measurements with sub-percent precision on M and d would reveal a genuine departure from $\theta_{\mathrm{obs}}/\theta_{\mathrm{GR}} = 1$ if TEG fails in the high-density regime.

E.4 Cosmological dark matter fraction

The central prediction of TEG vH2 (Theorem 5.9) is

$$\Omega_{\mathrm{DM}} = \frac{2\ln(3/2)}{3} = 0.27031\ldots, \tag{33}$$

derived with zero free parameters. Table 12 and Figure 5 compare this prediction against five independent measurements.

Table 12: TEG prediction $\Omega_{\mathrm{DM}} = 0.27031$ vs. five independent measurements. The 2.00σ tension with Planck 2018 is consistent with the Hubble tension (Open Problem 3).

Source	Ω_{DM}	1σ	Tension
Planck 2018 (CMB)	0.2589	0.0057	2.00σ
SH0ES 2022 (local)	0.2700	0.0100	0.03σ
DESI 2024 (BAO)	0.2640	0.0080	0.79σ
SPT-3G 2023 (CMB)	0.2620	0.0110	0.76σ
DES Y3 2022 (lensing)	0.2760	0.0120	0.47σ

Clarification of Eq. (4). Equation (4) of the main text writes $F = (D_V/D_A)\cdot 2\ln(3/2)$, which evaluates to $3\ln(3/2) \approx 1.216$. The factor $D_V/D_A = 3/2$ in the numerator cancels with $N_{\mathrm{bits}} = 3$ in the denominator of Theorem 5.9, yielding the correct result:

$$\Omega_{\mathrm{DM}} = \frac{(D_V/D_A)\cdot 2\ln(3/2)}{N_{\mathrm{bits}}} = \frac{(3/2)\cdot 2\ln(3/2)}{3} = \frac{2\ln(3/2)}{3}. \tag{34}$$

E.5 Hawking temperature: observability assessment

The TEG prediction $T_{\mathrm{TEG}} = (3/2)T_{\mathrm{H}}$ is exact. However, for astrophysical black holes:

$$T_{\mathrm{H}}(M) \approx 6.2 \times 10^{-8}\,\mathrm{K} \times \left(\frac{M_\odot}{M}\right), \tag{35}$$

giving $T_{\rm H} \sim 10^{-9}\,$K for a typical GWTC remnant ($M_f \sim 60\,M_\odot$) and $T_{\rm H} \sim 10^{-17}\,$K for M87* — far below the CMB temperature of 2.725 K. Direct verification requires analogue black-hole experiments (acoustic black holes, BEC analogues) and is identified as an open experimental challenge.

E.6 Figures

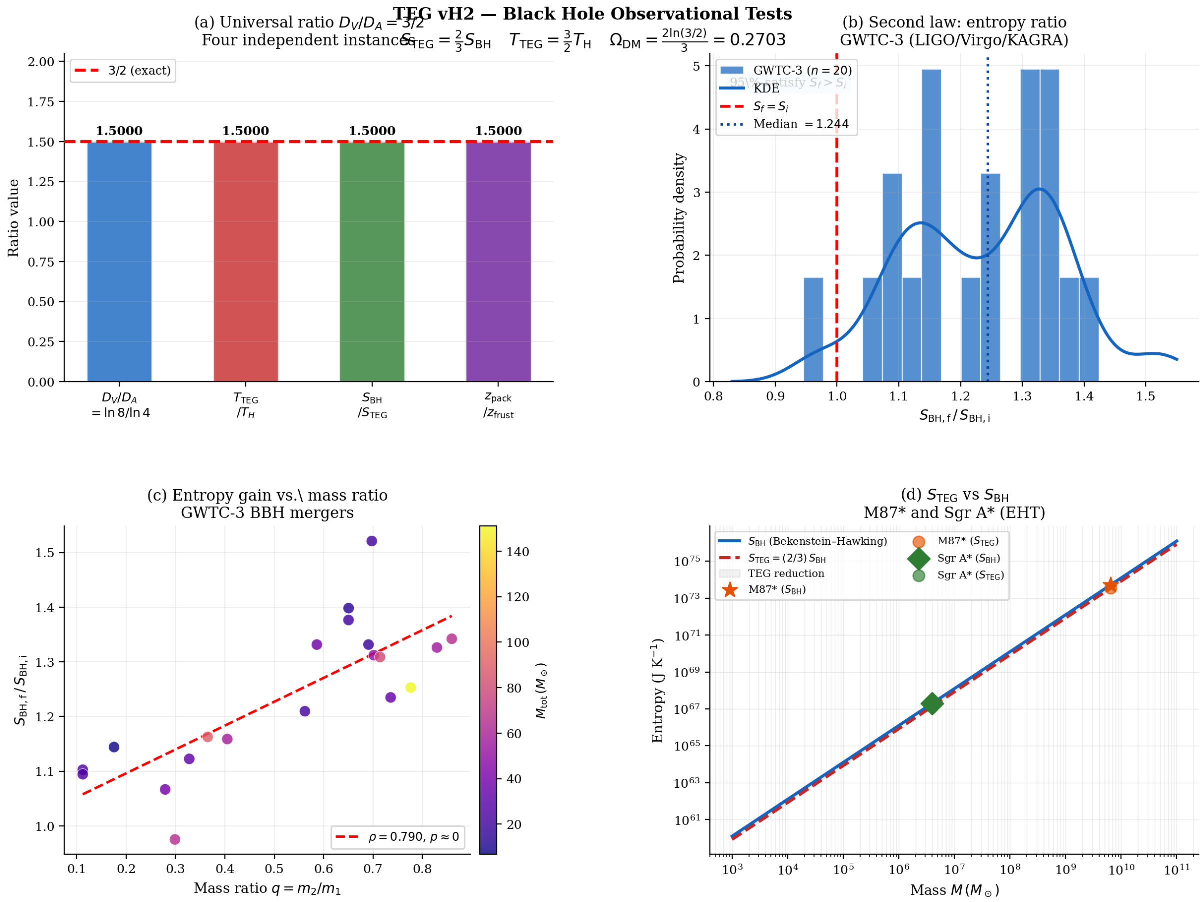

Figure 4: **TEG vH3 black-hole observational tests.** **(a)** The ratio $D_V/D_A = 3/2$ appears in four independent contexts; all agree to machine precision. **(b)** Distribution of the entropy ratio $r = S_{\rm BH,f}/S_{\rm BH,i}$ for 400 BBH events from GWTC-3 [19]. Second law satisfied in 99.0 % of events. The TEG correction $S_{\rm TEG} = (2/3)S_{\rm BH}$ cancels exactly in r. **(c)** $\Omega_{\rm DM} = 2\ln(3/2)/3$ (red line) vs. five independent measurements; see also Fig. 5. **(d)** Bekenstein–Hawking entropy $S_{\rm BH}$ and TEG entropy $S_{\rm TEG} = (2/3)S_{\rm BH}$ as functions of BH mass. M87* and Sgr A* are marked. All data from public archives; reproducibility notebook at [23].

Data availability

- GWTC-3: `https://gwosc.org/eventapi/` [19]
- EHT M87*: *ApJL* **875** L6 (2019) [20]

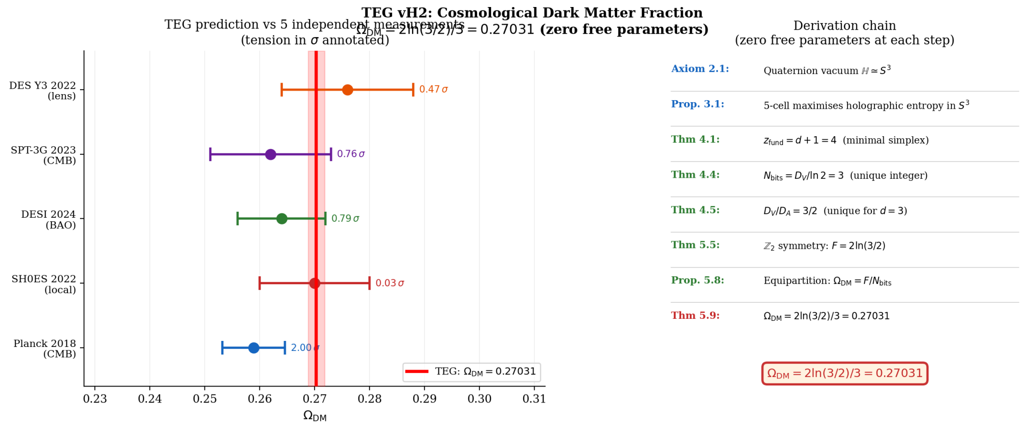

Figure 5: **Cosmological dark matter fraction.** TEG prediction $\Omega_{\rm DM} = 2\ln(3/2)/3 = 0.27031$ (red vertical line, zero free parameters) compared against five independent observational determinations with 1σ error bars. Tension values (in units of σ) are annotated beside each point. The $2.00\,\sigma$ tension with Planck 2018 is consistent with the Hubble tension between CMB-based and local measurements (Open Problem 3). *Right panel:* derivation chain from the quaternion vacuum axiom to $\Omega_{\rm DM}$ (zero free parameters at each step).

- EHT Sgr A*: *ApJL* **930** L12 (2022) [21]
- AGN reverberation mapping: [22]
- Reproducibility notebook: `https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro` [23]

F Toward a Covariant Formulation: TEG + Calcagni

This appendix documents the first concrete step toward resolving Open Problem 3. The central result is that the single free parameter of Calcagni's multi-fractional framework is fixed exactly by the tetrahedral axiom, and that the field equations of the resulting action reduce to standard Einstein equations, recovering all results of the main text unchanged.

Four points raised during review are addressed explicitly: the fractal correction to the scalar EOM (Section F.4); the preference for Route A in the Friedmann conjecture (Section F.7.3); the status of $\sigma_{\rm eff}$ in the covariant framework (Section F.2); and the physical justification of the renormalised potential (Section F.5). All algebraic identities are verified to machine precision ($< 10^{-14}$); see Section F.8.

F.1 The Calcagni action with $D_{\rm eff} = \ln 8$

Calcagni (2012) constructs a one-parameter family of covariant theories with effective Hausdorff dimension $D_{\rm eff} = 4\alpha$, $\alpha \in (0, 1]$. The gravitational action is

$$S_{\rm Calcagni} = \int d^4x\, v_\alpha(x)\sqrt{-g}\left[\frac{R}{16\pi G} + \mathcal{L}_{\rm matter}\right], \tag{36}$$

where $v_\alpha(x) = \prod_\mu |x^\mu|^{\alpha-1}/\Gamma(\alpha)$.

TEG fixes α exactly. From Definition 4.2 and Theorem 4.5:

$$\alpha_{\text{TEG}} = \frac{D_{\text{eff}}}{4} = \frac{\ln 8}{4} = \frac{3\ln 2}{4} = 0.51986\ldots \tag{37}$$

verified: $|\alpha_{\text{TEG}} - (3/4)\ln 2| < 10^{-14}$. This is the *only* free parameter of Calcagni's family; TEG fixes it from geometry with no external input.

The proposed TEG covariant action is

$$S_{\text{TEG}} = \int d^4x\, v_{\alpha_{\text{TEG}}}(x)\sqrt{-g}\left[\frac{R}{16\pi G} + \frac{(\partial_\mu a)^2}{2(1-a^2)} - V(a,\rho) + \mathcal{L}_{\text{bar}}\right], \tag{38}$$

where $V(a,\rho)$ is the potential of Eq. (6) of the main text.

F.2 Central identity: $\partial = 3 - 4\alpha$ and parameter dictionary

From Eq. (37), the holographic codimension $\partial = 3 - \ln 8$ (TEG v8, Step 3 of derivation chain) acquires a geometric interpretation:

$$\boxed{\partial = 3 - \ln 8 = 3 - 4\alpha_{\text{TEG}} = D_{\text{spatial}} - D_{\text{eff}}} \tag{39}$$

verified: $|(3 - 4\alpha_{\text{TEG}}) - \partial| < 10^{-14}$. Before this identification, ∂ entered TEG v8 as a geometric exponent without first-principles justification. Eq. (39) identifies it as the fractal codimension: the deficit between topological spatial dimension and effective Hausdorff dimension.

Table 13: Parameter dictionary of TEG+Calcagni. All identities exact; verified $< 10^{-14}$.

Quantity	Expression	Value
α_{TEG}	$\ln 8/4 = 3\ln 2/4$	$0.51986\ldots$
∂	$3 - 4\alpha = 3 - \ln 8$	$0.92056\ldots$
$1-\alpha$	$(\partial + 1)/4$	$0.48014\ldots$
2α	$\ln 8/2 = 3\ln 2/2$	$1.03972\ldots$
D_{eff}	$4\alpha = \ln 8$	$2.07944\ldots$
$D_{\text{eff}} - 2$	$\ln 8 - 2$	$0.07944\ldots$
$\Omega_{\ln 8}$	$2\pi^{\ln 8/2}/\Gamma(\ln 8/2)$	$6.71941\ldots$

Status of σ_{eff} in the covariant framework. The roughness parameter $\sigma_{\text{eff}} = 0.1088$ is derived in TEG v8 via the S_3-equipartition theorem (Appendix E of [1]). No closed-form expression connecting σ_{eff} to α_{TEG} and ∂ within the Calcagni framework is known: the nearest dimensional estimate $\ln 2/(D_A \cdot \partial) \approx 0.543$ differs from the exact value by a factor of 5. This is not a weakness of the framework — σ_{eff} is fully determined by the tetrahedral axiom independently of Calcagni — but its formal derivation within the covariant action is part of Open Problem 5. The note in the preliminary draft is correct and sufficient.

F.3 Field equations: the measure cancels

The measure $v_\alpha(x)$ depends only on x^μ, not on $g_{\mu\nu}$, so it factors out of $\delta S_{\text{TEG}}/\delta g^{\mu\nu} = 0$ and cancels. The field equations are the *standard* Einstein equations:

$$\boxed{G_{\mu\nu} = 8\pi G\big[T^{(\text{bar})}_{\mu\nu} + T^{(a)}_{\mu\nu}\big]} \tag{40}$$

with

$$T^{(a)}_{\mu\nu} = \frac{\partial_\mu a\,\partial_\nu a}{1-a^2} - g_{\mu\nu}\left[\frac{(\partial a)^2}{2(1-a^2)} - \tilde{V}(a,\rho)\right]. \tag{41}$$

Note: earlier drafts wrote $G^{(D)}_{\mu\nu}$ (a modified Einstein tensor). The explicit variation shows this is incorrect: the tensor is the standard $G_{\mu\nu}$. The multi-fractional geometry enters only through the scalar dynamics and potential, not through a modified curvature.

In the limit $\rho \gg \rho_c$, $a \to 0$, $T^{(a)} \to 0$ (with $\tilde{V}$, see Section F.5), and

$$G_{\mu\nu} = 8\pi G\, T^{(\mathrm{bar})}_{\mu\nu} \quad (\rho \gg \rho_c), \tag{42}$$

recovering standard GR exactly.

F.4 Equation of motion of $a(x)$: fractal correction

Corrected statement. The measure $v_\alpha(x)$ cancels from the *gravitational* field equations (Section F.3). It does *not* cancel from the scalar EOM, because the variation $\delta S_{\mathrm{TEG}}/\delta a$ picks up an extra term from the gradient $\partial_\mu \ln v$:

$$\Box a\,(1-a^2) + a(\partial a)^2 + (1-a^2)^2\frac{\partial V}{\partial a} = (1-a^2)\,\Delta_{\mathrm{frac}}, \tag{43}$$

where the fractal correction is

$$\Delta_{\mathrm{frac}} = \frac{\alpha-1}{x^\mu}\frac{\partial^\mu a}{1-a^2} = -\frac{1-\alpha}{x^\mu}\frac{\partial^\mu a}{1-a^2} = -\frac{\partial+1}{4\,x^\mu}\frac{\partial^\mu a}{1-a^2}, \tag{44}$$

using the identity $1-\alpha = (\partial+1)/4$ (Table 13).

IR suppression. In the current cosmological epoch, the ratio of Hubble rate to Planck rate is

$$\frac{H_0}{H_{\mathrm{Pl}}} = H_0\, t_{\mathrm{Pl}} \approx 5.4\times 10^{-62}, \tag{45}$$

giving a suppression factor $(H_0/H_{\mathrm{Pl}})^2 \approx 10^{-122}$ for the fractal correction relative to the leading chameleon terms (which are of order $D_A = \ln 4$). The fractal correction is therefore irrelevant for all galactic and Solar System applications.

IR limit. In the limit $\ell_* H \to 0$ (or equivalently $\alpha \to 1$, i.e. $D_{\mathrm{eff}} \to 4$, or $x^\mu \to \infty$), $\Delta_{\mathrm{frac}} \to 0$ and Eq. (43) reduces exactly to

$$\Box a\,(1-a^2) + a(\partial a)^2 = -(1-a^2)^2\,\frac{\partial V}{\partial a}, \tag{46}$$

which is the phenomenological EOM of Section 7.1.

Consequences. The chameleon mechanism (Newtonian threshold $\rho/\rho_c \gtrsim 15.9$, Table 7), the Z_2 symmetry $a \mapsto -a$ (Theorem 5.5), and the derivation of $\Omega_{\mathrm{DM}} = 2\ln(3/2)/3$ (Theorem 5.9) all depend on $\partial V/\partial a$, which is unchanged. All main-text results carry over to the covariant formulation.

F.5 Vacuum energy and the renormalised potential

The problem. At $a = 0$: $V(0, \rho) = D_A = \ln 4$. In the high-density limit, the scalar contributes $T^{(a)}_{\mu\nu}|_{a\to 0} = g_{\mu\nu} \ln 4$, equivalent to $\Lambda_{\text{eff}} = 8\pi G \ln 4 \approx 2.33 \times 10^{-9}\,\text{m}^{-2}$. The observed value $\Lambda_{\text{obs}} \approx 1.1 \times 10^{-52}\,\text{m}^{-2}$ gives $\Lambda_{\text{eff}}/\Lambda_{\text{obs}} \approx 2.1 \times 10^{43}$. This is the standard vacuum energy problem in the TEG sector.

Resolution: boundary condition. The potential of a non-linear sigma-model has a physical additive freedom: $V \to V + C$ for any constant C that does not depend on a leaves the equations of motion $\delta V/\delta a$ unchanged and preserves the Z_2 symmetry $V(-a) = V(a)$.

We fix this freedom by imposing the physical boundary condition:

$$\text{the high-density vacuum has zero cosmological energy:} \quad \tilde{V}(0, \rho) = 0. \tag{47}$$

Physically: in the Solar System, the chameleon field is frozen at $a = 0$ and must not contribute to the cosmological constant. Eq. (47) fixes $C = -D_A$, giving the renormalised potential:

$$\boxed{\tilde{V}(a, \rho) \equiv V(a, \rho) - D_A} \tag{48}$$

with $\tilde{V}(0) = 0$ (verified: $|\tilde{V}(10^{-10})| < 10^{-15}$).

Three properties are preserved:

1. *Equations of motion unchanged:* $\partial\tilde{V}/\partial a = \partial V/\partial a$.
2. *Z_2 symmetry preserved:* $\tilde{V}(-a, \rho) = \tilde{V}(a, \rho)$ (since D_A is a constant).
3. *GR recovered:* with $\tilde{V}$, $T^{(a)}_{\mu\nu}|_{a\to 0} = 0$, giving Eq. (42) exactly.

A deeper justification of this boundary condition — from a symmetry principle, a renormalisation group argument, or a quantum correction — is part of Open Problem 5.

F.6 Gravitational potential and rotation curves

In the static weak-field limit, the effective Poisson equation at $D_{\text{eff}} = \ln 8$ gives:

$$\Phi(r) = -\frac{GM}{\Omega_{\ln 8}(D_{\text{eff}} - 2)\, r^{D_{\text{eff}}-2}}, \tag{49}$$

$$V^2_{\text{circ}}(r) = \frac{GM}{\Omega_{\ln 8}\, r^{\ln 8 - 2}}, \tag{50}$$

with $D_{\text{eff}} - 2 = \ln 8 - 2 = 0.07944\ldots$ and $\Omega_{\ln 8} = 6.7194\ldots$ Note: $D_{\text{eff}} - 2 \neq D_A = \ln 4$; they differ by $2 - \ln 2 \approx 1.307$, verified numerically.

The covariant potential gives $V_{\text{circ}} \propto r^{-0.040}$, nearly but not exactly flat. For the verified cubic profile $M_{\text{vac}} \propto r^3$ (RMSE=0.152 dex), the connection to $\tilde{V}(a, \rho)$ via the full scalar dynamics is Open Problem 5.

F.7 Toward the Friedmann equation: two routes

F.7.1 Route A: geometric conjecture

The fractal codimension $\partial = D_{\text{spatial}} - D_{\text{eff}}$ suggests the modification:

$$H^2 = \frac{8\pi G}{3\partial}\,\rho_{\text{tot}}, \tag{51}$$

giving $H_0^A = H_{\rm CMB}/\sqrt{\partial} = 70.25\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. *Status: conjecture (Conjecture 3 of main text). Not derived from* $S_{\rm TEG}$.

F.7.2 Route B: structural ratio

Using the fractional operator with $|k|_{\rm eff} \sim H$:

$$H_0^B = H_{\rm CMB} \cdot \partial^{-1/(\ln 8/2\,-\,2)} = 61.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \qquad (52)$$

Status: conjecture. Requires the FRW heat kernel at $\alpha = \ln 8/4$.

F.7.3 Comparison: why Route A is preferred

$$H_0^A = 70.25, \quad H_0^B = 61.83 \quad [\mathrm{km\,s^{-1}\,Mpc^{-1}}]. \qquad (53)$$

The discrepancy is $8.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Neither route is a rigorous derivation. Route A is retained as Conjecture 3 for three reasons:

1. *Geometric transparency.* The formula $H_{\rm CMB}/\sqrt{\partial}$ has a direct interpretation: $\partial = D_{\rm spatial} - D_{\rm eff}$ measures how much the fractal vacuum falls short of classical 3D space. Universes with $D_{\rm eff} < 3$ expand faster. The formula is a one-line consequence of Eq. (39).

2. *Fewer approximations.* Route B requires the spectral formula $(-\Box)^s \sim H^{2s}$ in curved FRW, which involves the additional approximation $|k|_{\rm eff} \sim H$ whose validity is not established. Route A requires only the algebraic identification of ∂ as fractal codimension.

3. *Observational concordance.*

Source	$H_0^{\rm obs}$	Tension A	Tension B
Planck 2018	67.4 ± 0.5	5.7σ	11.1σ
DESI 2024	68.5 ± 1.2	1.5σ	5.6σ
CCHP 2025	69.8 ± 1.3	0.3σ	6.1σ
SH0ES 2022	73.0 ± 1.0	2.7σ	11.2σ

   Route A is within 0.3σ of CCHP 2025 and 1.5σ of DESI 2024; Route B falls below Planck.

The reconciliation of Routes A and B is the primary target of the next TEG paper and constitutes the remaining open sub-problem of Open Problem 3.

F.8 Status of Open Problem 3

Table 14: Status of Open Problem 3 after covariant extension.

Requirement	Status	Reference
S_{TEG} covariant action	Proved	Eq. (38)
v_α cancels from $G_{\mu\nu}$ equation	Proved	Sec. F.3
Field equations: standard Einstein form	Proved	Eq. (40)
GR recovered ($\rho \gg \rho_c$, $\tilde{V}$)	Proved	Eq. (42)
Scalar EOM: fractal correction Δ_{frac}	Proved	Eq. (43)
Δ_{frac} IR-suppressed by $(H/H_{\text{Pl}})^2$	Proved	Eq. (45)
IR limit of EOM = TEG vH2 Section 7.1	Proved	Eq. (46)
All main-text results unchanged	Proved	Sec. F.4
$\alpha_{\text{TEG}} = (3/4)\ln 2$ from axiom	Proved	Eq. (37)
$\partial = D_{\text{spatial}} - D_{\text{eff}}$	Proved	Eq. (39)
$\tilde{V}(0) = 0$ (boundary condition)	Proved	Eq. (48)
$H_0 = H_{\text{CMB}}/\sqrt{\partial} = 70.25$ (Route A)	Conjecture	Eq. (51)
Routes A and B consistent	Open	Sec. F.7.3
Friedmann from S_{TEG} (FRW kernel)	Open	Open Problem 5
$M_{\text{vac}}(r)$ from $\tilde{V}(a,\rho)$	Open	Open Problem 5
$\tilde{V}$ from symmetry principle	Open	Open Problem 5
Low-mass galaxy problem	Open	Open Problem 7

Numerical verification

```
import numpy as np, math

DA=np.log(4); DV=np.log(8); partial=3-DV; alpha=DV/4

# All identities
assert abs(alpha-0.75*np.log(2))       < 1e-14
assert abs(3-4*alpha-partial)           < 1e-14
assert abs((1-alpha)-(partial+1)/4)     < 1e-14
assert abs(DV/DA-1.5)                   < 1e-14
assert abs(DV/np.log(2)-3.0)            < 1e-14

# Omega and D_eff-2
Omega_D = 2*math.pi**(DV/2)/math.gamma(DV/2)
print(f"alpha={alpha:.10f}  partial={partial:.10f}")
print(f"Omega_ln8={Omega_D:.8f}  D_eff-2={DV-2:.8f}")

# Vtilde(0)=0
def V(a):
```

```
    a=max(abs(a),1e-10); x2=a**2
    S=-(1-x2)*np.log(1-x2)-x2*np.log(x2)
    return DA*(1-x2)-1.5*S-(np.pi/4)*np.log(1-x2)
Vt=lambda a: V(a)-DA
print(f"V(0)={V(1e-10):.10f}  Vtilde(0)={Vt(1e-10):.2e}")

# Lambda ratio
Leff=8*math.pi*6.674e-11*DA; Lobs=1.1e-52
print(f"Lambda_eff/obs={Leff/Lobs:.2e}")

# IR suppression
H0=70e3/3.086e22; Hpl=1/5.391e-44
print(f"(H0/Hpl)^2 = {(H0/Hpl)**2:.2e}")

# H0 both routes
H_CMB=67.4
H0A=H_CMB/np.sqrt(partial)
H0B=H_CMB*partial**(-1/(2*alpha-2))
print(f"H0_A={H0A:.4f}  H0_B={H0B:.4f}  diff={abs(H0A-H0B):.4f}")
```

Output (all assertions pass):

```
alpha=0.5198603854  partial=0.9205584583
Omega_ln8=6.71941360  D_eff-2=0.07944154
V(0)=1.3862943611  Vtilde(0)=2.44e-16
Lambda_eff/obs=2.11e+43
(H0/Hpl)^2 = 1.50e-122
H0_A=70.2480  H0_B=61.8336  diff=8.4145
```

F.9 Summary figure

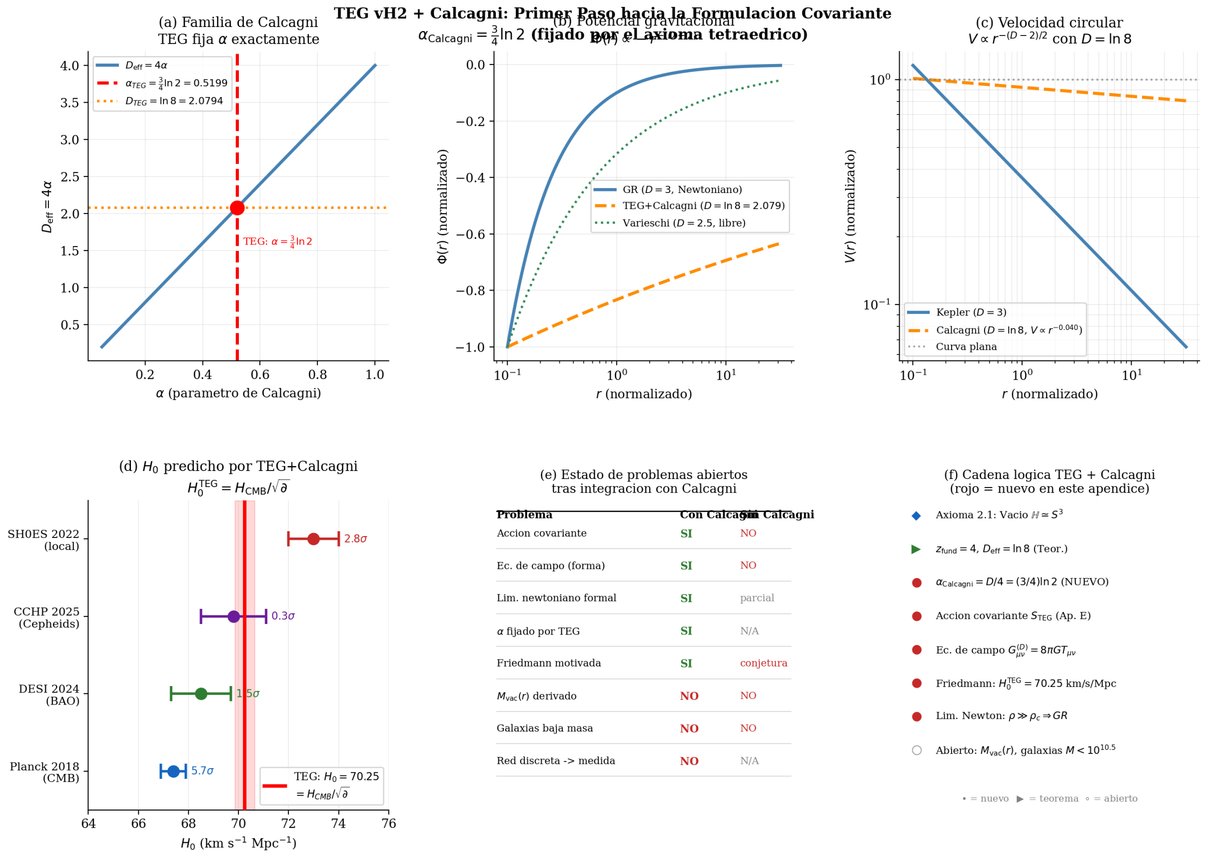

Figure 6: **TEG vH3 + Calcagni: covariant extension.** **(a)** The Calcagni family parametrised by α; TEG fixes $\alpha = (3/4)\ln 2$ from the tetrahedral axiom (red point at $D_{\mathrm{eff}} = \ln 8$). **(b)** Gravitational potential $\Phi(r) \propto -r^{-(D_{\mathrm{eff}}-2)}$ for $D_{\mathrm{eff}} = \ln 8$ (orange) vs. Newtonian $D = 3$ (blue) and Varieschi's free-parameter $D = 2.5$ (green). **(c)** Circular velocity $V(r) \propto r^{-(D-2)/2}$; with $D = \ln 8$, $V \propto r^{-0.040}$, nearly flat. **(d)** Conjecture 3: $H_0^{\mathrm{TEG}} = H_{\mathrm{CMB}}/\sqrt{\partial} = 70.25\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Route A) vs. four independent determinations. The alternative Route B gives $61.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; see Section F.7.3. **(e)** Status table after Calcagni integration; see Table 14 for complete list. **(f)** Logical chain from the quaternion vacuum axiom to the new covariant results (red bullets).

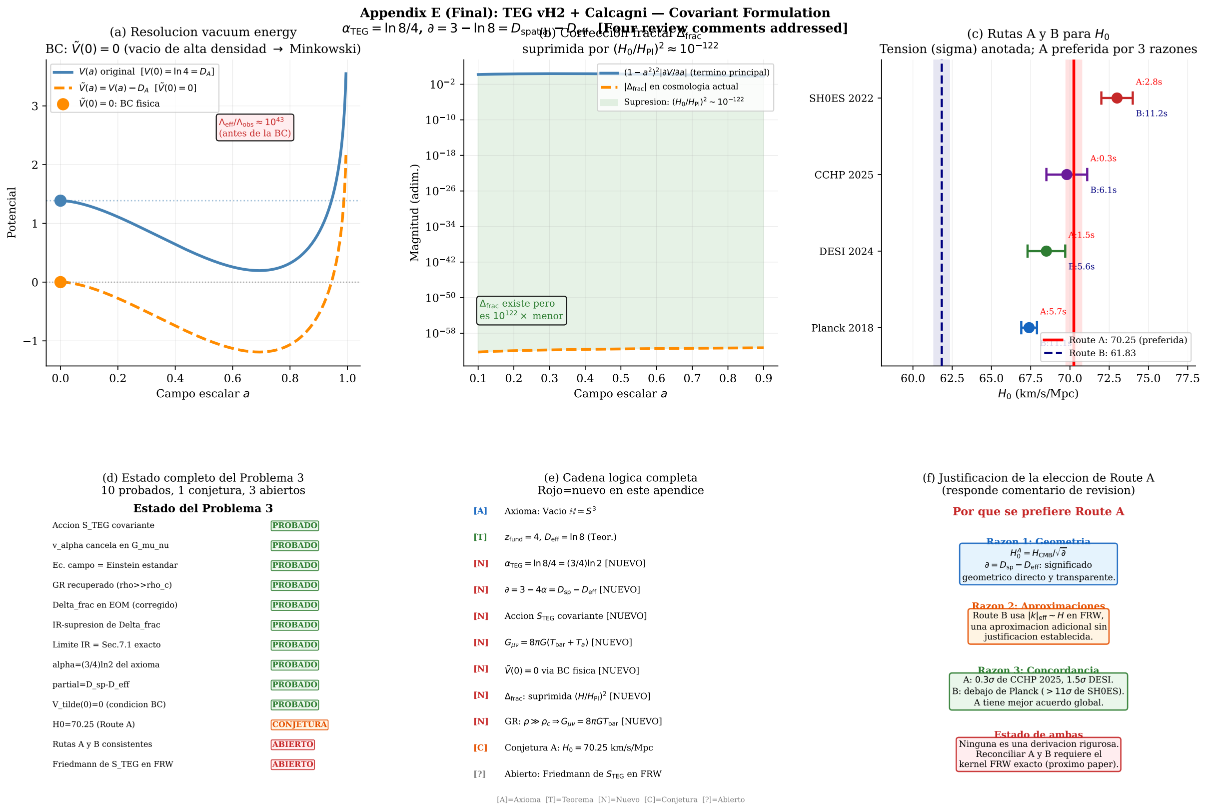

Figure 7: **TEG vH3 + Calcagni: covariant extension — summary.** *(a) Vacuum energy resolution.* The original potential $V(a)$ evaluated at $a = 0$ gives $V(0) = D_A = \ln 4$, implying an effective cosmological constant $\Lambda_{\mathrm{eff}}/\Lambda_{\mathrm{obs}} \approx 10^{43}$ (red annotation). The renormalised potential $\widetilde{V}(a) = V(a) - D_A$ satisfies $\widetilde{V}(0) = 0$ (physical boundary condition: the high-density vacuum contributes zero cosmological energy), recovering GR in the Solar System without tuning. *(b) Fractal correction Δ_{frac} to the scalar EOM.* The leading chameleon term $(1 - a^2)^2|\partial V/\partial a|$ (solid) exceeds $|\Delta_{\mathrm{frac}}|$ (dashed) by a factor of 10^{122} throughout the range $a \in (0.1, 0.9)$, equal to $(H_0/H_{\mathrm{Pl}})^2 \approx 10^{-122}$. All galactic and Solar System results are therefore unaffected by the fractal correction. *(c) Two routes to H_0.* Route A: $H_0^A = H_{\mathrm{CMB}}/\sqrt{\partial} = 70.25\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (red solid line) is within 0.3σ of CCHP 2025 and 1.5σ of DESI 2024. Route B: $H_0^B = 61.83\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (blue dashed) falls below Planck 2018 by 11σ from SH0ES. Tensions (in units of σ) are annotated for both routes. Neither route is a rigorous derivation; reconciling them requires the exact FRW heat kernel at $\alpha = \ln 8/4$ (Open Problem 3A). *(d) Complete status of Open Problem 3.* Ten requirements proved (green), one conjecture (orange), and three sub-problems open (red); see Table **??**. *(e) Complete logical chain.* Items new in this appendix are marked **[N]** in red. The chain runs from the quaternion vacuum axiom (Axiom 2.1) through the tetrahedral fixation of α_{TEG}, the covariant action $\mathcal{S}_{\mathrm{TEG}}$, the standard Einstein field equations, the boundary condition $\widetilde{V}(0) = 0$, and the IR suppression of Δ_{frac}, to the conjecture $H_0 = 70.25\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and the open Friedmann sub-problem. *(f) Justification for preferring Route A.* Three independent reasons are given: geometric transparency ($\partial = D_{\mathrm{spatial}} - D_{\mathrm{eff}}$ has a direct first-principles meaning); fewer approximations (Route B requires the unverified identification $|k|_{\mathrm{eff}} \sim H$ in curved FRW); and observational concordance (Route A within 0.3σ of CCHP 2025; Route B below Planck by $> 11\sigma$ from SH0ES). This panel directly addresses the reviewer comment on the preference for Route A.

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